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May 30, 2013

To: Members of the Executive Board

From: The Secretary

Subject: **Ghana—Staff Report for the 2013 Article IV Consultation**

Attached for consideration by the Executive Directors is the staff report for the 2013 Article IV consultation with Ghana, which is tentatively scheduled for discussion on **Wednesday, June 12, 2013**. Unless an objection from the authorities of Ghana is received prior to the conclusion of the Board's consideration, the document will be published. Any requests for modifications for publication are expected to be received two days before the Board concludes its consideration.

Questions may be referred to Ms. Daseking, AFR (ext. 37340), Mr. Baldini, ICD (ext. 38794), and Mr. Arze del Granado, AFR (ext. 39642).

Unless the Documents Section (ext. 36760) is otherwise notified, the document will be transmitted, in accordance with the procedures approved by the Executive Board and with the appropriate deletions, to the WTO Secretariat on Friday, June 7, 2013; and to the African Development Bank, the European Commission, and the Organisation for Economic Cooperation and Development, following its consideration by the Executive Board.

This document, together with a supplement providing an informational annex, will shortly be posted on the extranet, a secure website for Executive Directors and member country authorities. The supplement, which is not being distributed in hard copy, will also be available in the Institutional Repository; a link can be found in the daily list (<http://www-int.imf.org/depts/sec/services/eb/dailydocumentsfull.htm>) for the issuance date shown above.

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STAFF REPORT FOR THE 2013 ARTICLE IV CONSULTATION

May 29, 2013

KEY ISSUES

Near-term outlook: Strong growth momentum, but significant stability risks. GDP growth of 8 percent in 2012 is projected to continue into 2013, with inflation back above 10 percent. Ghana's strong democratic institutions and favorable prospects for oil and gas are attracting significant foreign direct investment (FDI). Yet, a large current account deficit, thin external buffers, and rising public debt expose the economy to serious risks. Energy sector problems could curtail growth, while heavy government borrowing is raising the cost of credit to the private sector.

Economic policy agenda: Advancing middle-income status. Ghana has made great strides in reducing poverty and has recently reached lower middle-income status. The economy, however, still relies heavily on agriculture and natural resources, particularly for exports, and most jobs are in the informal sector. The newly elected government has adopted an ambitious transformation agenda, centered on economic diversification, shared growth and job creation, and macroeconomic stability.

Macroeconomic policy mix: Need for realignment. With a near tripling of the cash deficit relative to 2011 to almost 12 percent of GDP, fiscal policy was highly expansionary in 2012. Monetary policy was tightened with some delay to halt a rapid currency depreciation, but success in stabilizing the currency came at the cost of double digit real interest rates. The immediate policy priority is to safeguard stability by rebuilding fiscal and external buffers and reduce debt, while keeping a tight monetary policy stance. Decisive fiscal consolidation will, in due course, allow for a reduction in interest rates. Going forward, successful economic transformation will require a realignment of spending away from wages and subsidies toward investment in infrastructure, while structural fiscal reforms are needed to restore policy credibility and build institutional resilience to the political cycle.

Financial sector: Containing vulnerabilities and removing structural constraints. The banking system has grown steadily in assets and profitability, but high interest rates are pricing out profitable investment projects, while raising the risk of a renewed increase in nonperforming loans. The Bank of Ghana should consider requiring banks to hold higher minimum capital buffers against a future downturn, while addressing gaps and inconsistencies in the banking laws, and deepening cooperation with regional counterparts to improve the regulation and supervision of foreign banks.

GHANA

Approved By
**Michael Atingi-Ego
and Mark Flanagan**

Discussions were held in Accra during April 2–12, 2013. The staff team comprised Christina Daseking (head), Javier Arze del Granado, Alfredo Baldini, Kevin Wiseman (all AFR), Monique Newiak (SPR), and Samir Jahjah, (Resident Representative). Martin Fukac (RES) participated to advise the Bank of Ghana on improvements to their inflation-targeting framework. Susan George (MCM) and Cedric Mousset (World Bank) headed the advance mission for enhancing financial sector surveillance. Messrs. Mojarrad (ED) and Abradu-Otoo (OED) participated in the discussions. The mission met with President Mahama, Vice-President Amissah-Arthur, Finance Minister Terkper, Bank of Ghana Governor Wampah, other senior officials, members of parliament, and representatives of the private sector, think tanks, trade unions, and civil society. For the preparation of this report, Imededdine Jerbi and Jean Vibar provided research and administrative support, respectively.

CONTENTS

RECENT DEVELOPMENTS AND OUTLOOK: HIGH POTENTIAL BUT SHORT-TERM RISK ____ 4

POLICY DISCUSSIONS _____ 7

ECONOMIC POLICY AGENDA: ADVANCING MIDDLE-INCOME STATUS _____ 7

FISCAL POLICY: REALIGNING PRIORITIES _____ 8

MONETARY POLICY: LITTLE ROOM FOR EASING _____ 16

STAFF APPRAISAL _____ 23

BOXES

1. Assessing Reserve Adequacy in Ghana _____ 12
2. External Sustainability Assessment _____ 13
3. Six Years of Inflation Targeting in Ghana: Policy Lessons and Challenges _____ 19
4. Enhanced Financial Sector Surveillance: Main Conclusions _____ 22

FIGURES

1. Real Sector Indicators _____ 5
2. External Indicators _____ 6
3. Fiscal Indicators _____ 9
4. Monetary and Financial Indicators _____ 17

TABLES

1. Selected Economic and Financial Indicators, 2011–18	26
2A. Summary of Central Government Budgetary Operations, 2011–18 (in percent of GDP)	27
2B. Summary of Central Government Budgetary Operations, 2011–18 (in millions of GHc)	28
2C. Summary of Budgetary Central Government Operations, 2008–18 (in percent of GDP)	29
2D. Summary of Budgetary Central Government Operations, 2008–18 (millions of GHc)	30
3. Monetary Survey, 2011–18	31
4. Balance of Payments, 2011–18	32
5. Financial Soundness Indicators, 2007–12	33

APPENDIXES

I. Risk Assessment Matrix	34
II. Inclusive Growth Successes and Challenges	36
III. Public Investment, Growth, and Debt Sustainability	45
IV. Implementation of the 2011 FSAP Update-Key Policy Recommendations	49
V. Draft Public Information Notice	50

RECENT DEVELOPMENTS AND OUTLOOK: HIGH POTENTIAL BUT SHORT-TERM RISK

1. Peaceful elections in December 2012 confirmed Ghana's strong democratic credentials.

President Mahama's National Democratic Congress was reelected in December with a small margin in the popular vote, but a sizeable parliamentary majority. Uncertainty arises from the opposition's challenge of the election results in court. The government faces high expectations for transforming Ghana's nascent oil and gas resources into quality jobs and improved services and living conditions for its people.

2. Growth continued at a robust pace amid rising fiscal and external imbalances. 2012 GDP growth of close to 8 percent was supported by expansionary fiscal policy, mirroring patterns of past election years. Fiscal pressures came to the fore in a rising public sector wage bill and costly energy subsidies that pushed the deficit close to 12 percent of GDP. The fiscal expansion also led to a significant deterioration in the public debt ratio and a widening current account deficit.

3. The policy mix was suboptimal. While fiscal policy became increasingly expansionary in the course of 2012, the Bank of Ghana (BoG) tightened monetary policy in the second half of the year to arrest a rapid depreciation of the cedi. The currency has since stabilized, with recent depreciations

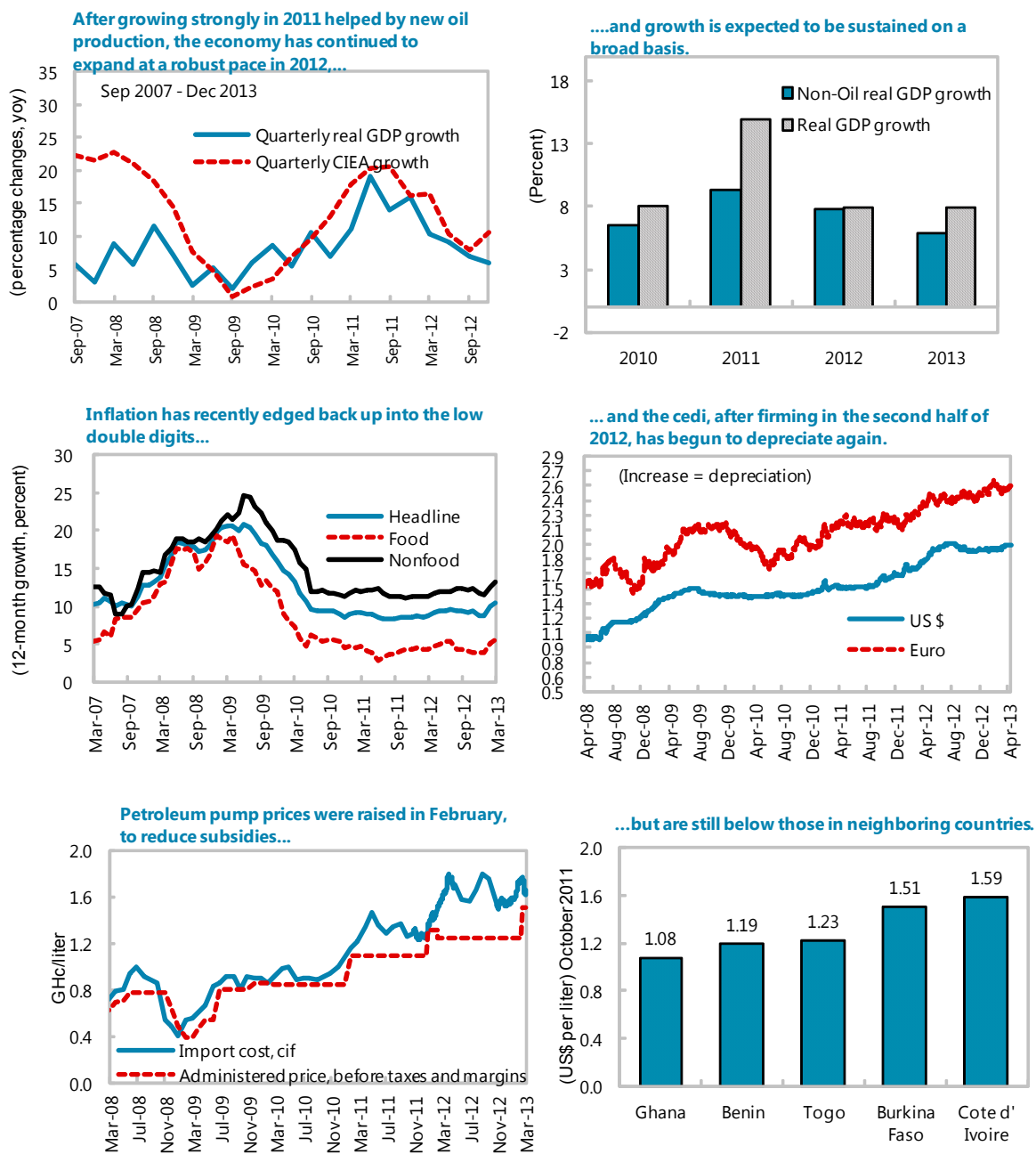
in line with inflation differentials, but at the cost of double-digit real interest rates. Although consumer price inflation stayed in the single digits in 2012, this outcome was helped by low food and repressed domestic fuel prices. With rising core inflation (excluding food and energy) and a February increase in fuel prices, inflation has moved back above 10 percent.

4. The growth momentum continues into 2013, with inflation on the rise. While non-oil growth is likely to decelerate, as a result of energy disruptions and high real interest rates, increased oil production is projected to keep overall growth close to 8 percent. Survey-based inflation expectations remain elevated at above 10 percent. The current account deficit is projected to stay high at 12 percent of GDP, despite a moderation in import growth, reflecting a weaker outlook for cocoa and gold exports. Staff projects a small reduction in the fiscal deficit (financing) to 10 percent of GDP this year, about 1 percent of GDP higher than the authorities' budget projections, mainly reflecting a continuation of energy subsidies.

Macroeconomic Indicators, 2012-2018							
	2012	2013	2014	2015	2016	2017	2018
	Prel.			Proj.			
(Annual percentage change)							
Real GDP	7.9	7.9	6.1	5.5	5.1	5.1	8.0
Nonoil real GDP	7.8	5.9	6.0	5.9	5.8	5.8	5.8
Inflation, end-of-period	8.8	10.8	8.8	7.7	7.3	7.1	6.9
Exchange rate (average, GhC per U.S. dollar)	17.1	8.7	8.0	6.5	5.7	5.2	5.0
(Percent of GDP; unless otherwise indicated)							
Overall fiscal balance (financing)	-11.8	-10.0	-8.8	-6.2	-6.2	-6.2	-6.2
Central Government debt (net)	48.0	49.5	50.5	50.5	50.7	52.6	52.1
Current account balance (percent of GDP)	-12.2	-11.9	-9.1	-8.6	-8.6	-8.6	-7.4
BOG GIR stock (US\$ mil)	5,349	4,927	5,645	6,579	7,407	8,324	10,139
Percent of prospective imports (G & S)	2.8	2.5	2.7	3.0	3.2	3.3	3.8

Sources: Ghanaian authorities and IMF staff projections.

Figure 1. Ghana: Real Sector Indicators

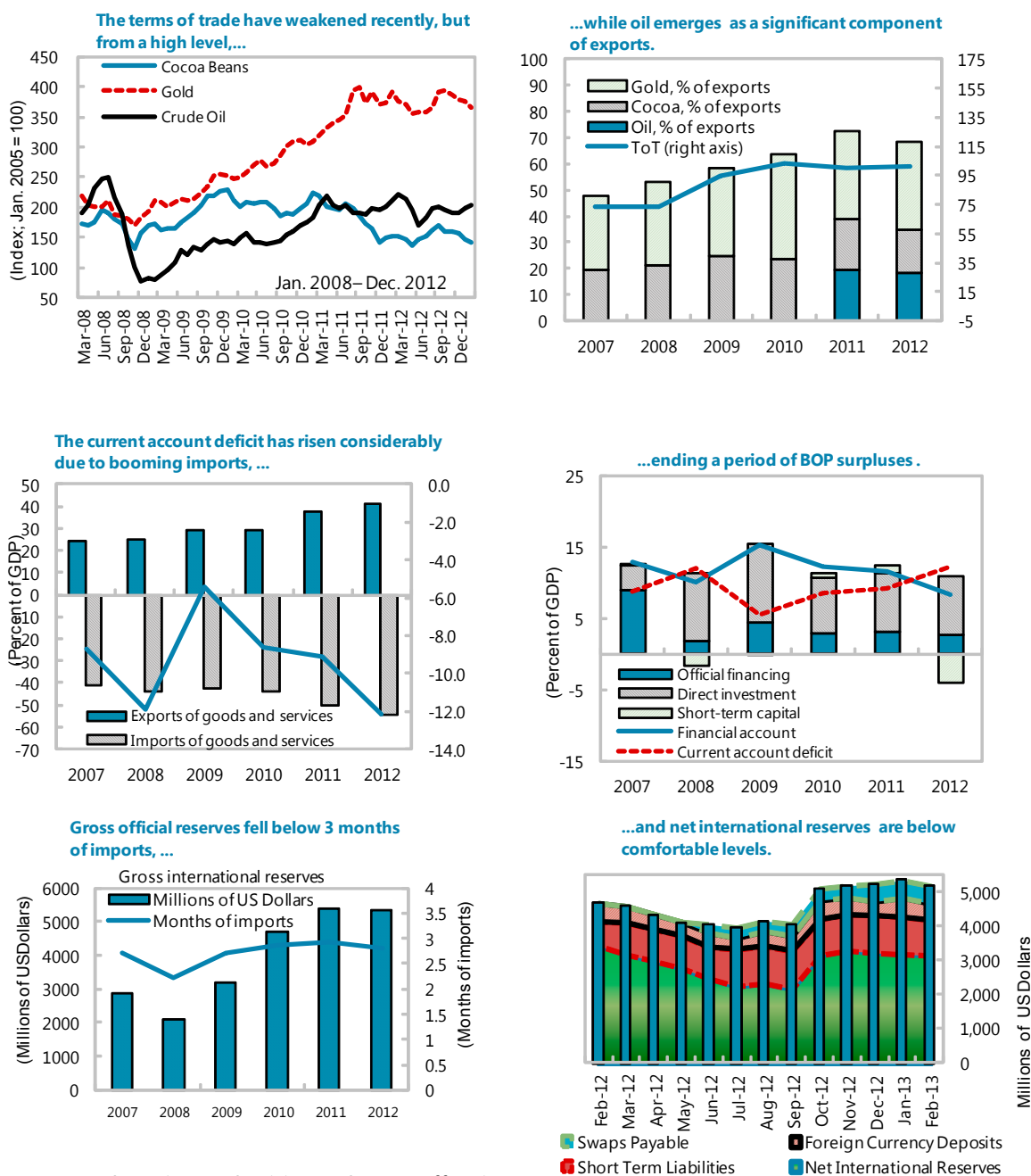


Source: Ghanaian authorities and IMF staff estimates and projections.

¹ The CIEA is the Bank of Ghana's composite index of real economic activity.

GHANA

Figure 2. Ghana: External Indicators



5. Ghana's medium-term outlook is positive, if vulnerabilities are addressed. The economy continues to attract significant FDI, aided by the prospect of rising oil and gas production. Non-oil growth is projected to stabilize at a still robust level of 5–6 percent. A planned reduction in the fiscal deficit to about 6 percent of GDP is feasible by 2015, if policies are implemented as envisaged, contributing to a gradual decline in inflation. The current path of fiscal consolidation together with increased oil and gas production from new fields would reduce the current account deficit to a still high level of 7½ percent of GDP by 2018.

6. Short-term stability risks are significant (Appendix I). Deteriorating public debt dynamics and low official reserve buffers make the economy susceptible to risks from delayed fiscal adjustment, weaker terms of trade, and rollover risks from foreign holdings of government bonds. Domestically, energy sector problems pose downside risks to growth and fiscal outcomes.

7. Recommendations from the 2011 Article IV consultation remain valid today. At the time, Executive Directors stressed the importance of stronger fiscal management and called for rigorous implementation of the authorities' plan to reduce the cash deficit, including the re-establishment of cost-recovery pricing of energy products and the avoidance of domestic arrears. Other recommendations included improvements in tax administration, payroll management, and debt management, supported by prudent borrowing strategies. While the government implemented a number of important measures in these areas, with impressive improvements particularly on the revenue side, the latest fiscal developments have substantively reversed the achievements of 2011.

POLICY DISCUSSIONS

ECONOMIC POLICY AGENDA: ADVANCING MIDDLE-INCOME STATUS

There was full agreement that successful realization of the government's growth agenda will require careful prioritization and sequencing, supported by fiscal consolidation.

8. Ghana is at a critical stage in its economic development. While having reached lower middle-income status, the economy still relies heavily on agriculture and natural resources and most jobs are in the informal sector. Similarly, the concentration of exports on three commodities (gold, cocoa, and oil) makes the economy vulnerable to terms-of-trade shocks. Although poverty and social indicators have improved significantly, about a quarter of the population still lives below the poverty line, and demographic trends will require the creation of 6–7 million jobs in the next 20 years to absorb new entrants into the labor market. Appendix II discusses Ghana's inclusive growth successes and challenges in more detail. It concludes that the government will need to leverage Ghana's strong governance and business environment and decisively tackle current growth constraints to diversify production toward manufacturing and higher-value agriculture. Better infrastructure, improved access to credit, further investment in health and education, and sustained macroeconomic stability will be central to ensure Ghana's continued robust and inclusive growth.

GHANA

9. The government's transformation agenda provided the context for the discussions. It pursues three broad objectives:

- *Economic diversification.* Leveraging oil and gas resources to create a robust job-creating manufacturing sector will require significant infrastructure investments and removal of the main bottlenecks to growth—inadequate and unreliable energy provision and lack of affordable private sector financing.
- *Social inclusion.* To make further advancements in poverty reduction, ensure that the benefits of growth are widely shared, and build a workforce ready to take on higher-skilled jobs, the government wants to further strengthen Ghana's social safety net and continue investments in utilities, health, and education, while improving the quality of social spending.
- *Macroeconomic and debt sustainability.* An updated debt sustainability analysis suggests that Ghana's risk of debt distress has risen, but remains moderate, provided the government successfully implements a sizeable fiscal adjustment (see supplement).

10. The government's immediate priority is to restore macroeconomic stability. Fiscal consolidation is needed to reduce existing vulnerabilities and allow for a gradual realignment in the policy mix in support of private investment and growth. At the same time, the government recognized that successful transformation requires administrative and institutional reforms to promote reliable policy implementation—including resilience to the political cycle—higher efficiency of public infrastructure investment, and deeper financial intermediation.

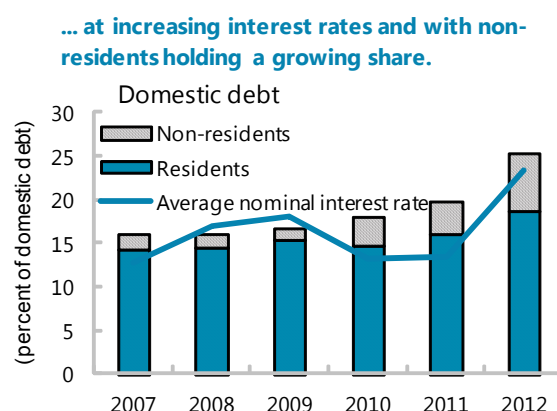
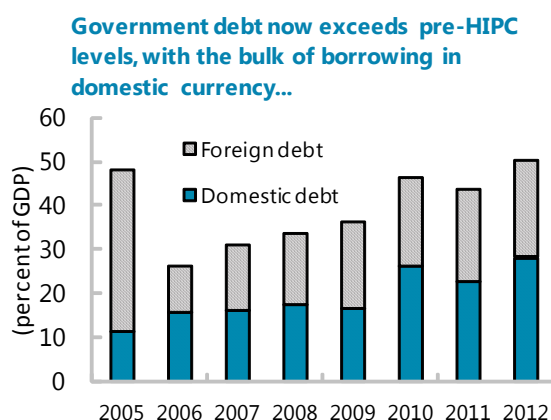
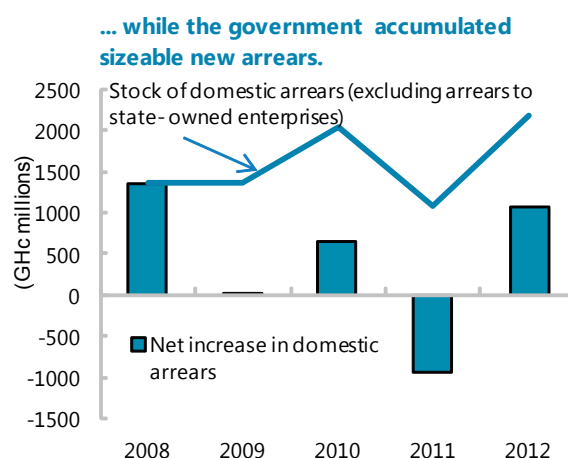
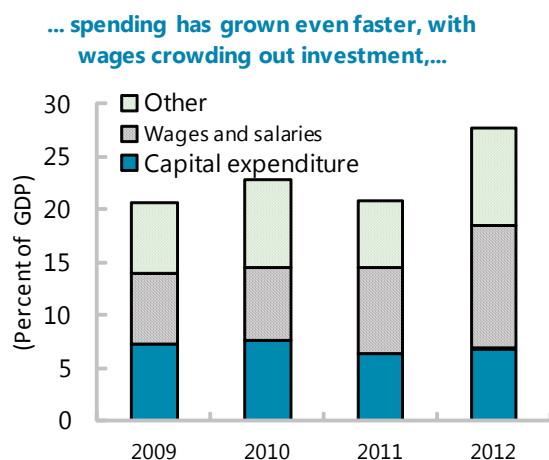
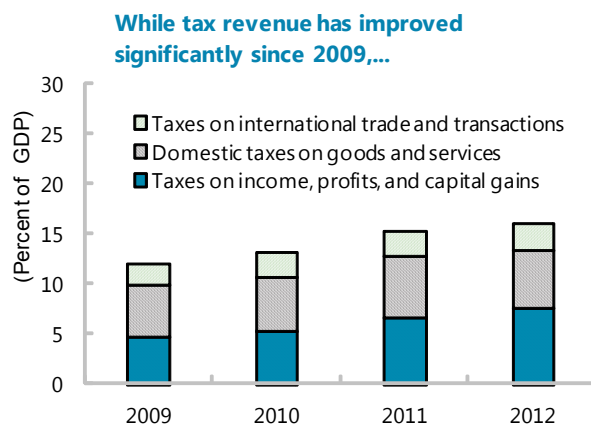
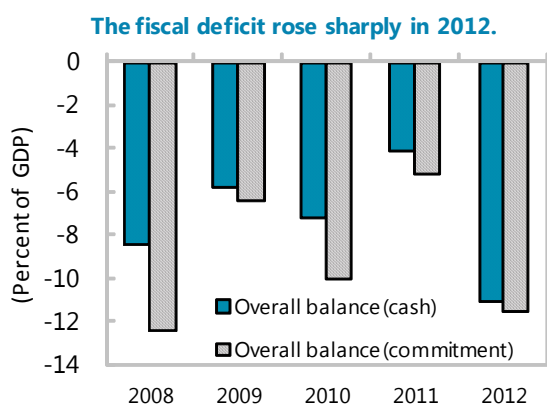
FISCAL POLICY: REALIGNING PRIORITIES

There was agreement that fiscal consolidation and a shift in the composition of spending from wages and subsidies to investment are needed to support growth, reduce external pressures, and keep debt sustainable. The mission strongly supported the government's plans for improving public financial management, while advocating more ambitious fiscal consolidation than currently planned.

11. The government's main policy task is to reverse the large fiscal slippages of 2012. The fiscal deficit (financing basis) rose to almost 12 percent of GDP—nearly twice the level anticipated in the revised budget (Figure 3). The government also accumulated net arrears of 2.8 percent of GDP, of which nearly half was to state-owned enterprises, mainly linked to the under-pricing of fuel and utilities. A shortfall in revenue and grants added to the deficit, but the main cause for the large fiscal deterioration was higher spending.

12. Failure to control the public wage bill was the key driver behind the larger deficit. Recurrent spending rose by 4¼ percentage points of GDP between 2011 and 2012, reflecting higher interest cost, energy subsidies and, in particular, a much larger wage bill, which grew by 47 percent

Figure 3. Ghana: Fiscal Indicators



Source: Ghanaian authorities and IMF staff estimates.

GHANA

in nominal terms. An 18 percent pay hike explains part of this increase, and new hiring appears to have offset the smaller-than-expected savings from the recent payroll audit, but many of the factors that could explain the increase have not yet been quantified. In addition, deferred wage payments from the single spine salary reform amounted to 2½ percent of GDP, twice the level included in the mid-year revised budget. The government has initiated a thorough examination of the payroll, realizing that its ability to control wages in the future hinges on its understanding of the significant overruns in 2012.

13. The mission saw difficulties in meeting the 2013 fiscal target in the absence of additional policy measures. The budget foresees a deficit of 9 percent of GDP on a cash basis, assuming a 12 percent nominal increase in the wage bill and higher tax revenue, mainly as a result of further efforts to strengthen the performance of the Ghana Revenue Authority (GRA). Other revenue measures—an audit and subsequent streamlining of VAT exemptions, an improved VAT bill (including VAT on financial services), and a windfall profit tax on mining—are expected to become effective only in 2014. Pressures on the expenditure side arise from the need to cover subsidies on fuel and utilities. While the former were significantly reduced with the February price increases, utility tariffs remain below cost-recovery levels. Without price adjustments, compensation to state-owned energy companies could exceed the budgeted amount by 1 percent of GDP by the end of the third quarter. Incorporating these additional expenditures, a delayed impact from the envisaged revenue measures, but also a higher nominal GDP (as a result of recent upward revisions to the 2012 data), staff projects a deficit of 10 percent of GDP this year.

14. The authorities acknowledged the challenges, but were positive that their deficit target could be achieved. They anticipated that the repair of the West African Gas Pipeline and new domestic power facilities would restore stable energy supply soon, which would then allow regular adjustments in utility tariffs. Spending control would be significantly strengthened with the full rollout of the Ghana Integrated Financial Management Information System (GIFMIS) and improved cash management, including closer scrutiny of monthly expenditure relative to forecasts. In response to revenue shortfalls in the first four months of the year, the authorities announced in late May their plans to submit new tax measures to parliament. These include the reintroduction of a stabilization levy—a temporary additional profit tax on certain sectors, such as financial services and mining which was in place also in 2009-11—levies on certain imports, increases in excise duties, a review of fees and charges, and administrative measures to raise GRA collections. The estimated combined impact is close to ½ percent of GDP for the six months starting in July, which would be broadly sufficient to offset the revenue shortfall through April. In addition, the government plans to undertake a major mid-term review of its policies in the summer, to assess the need for additional measures. The mission welcomed this review, stressing the urgency of reducing vulnerabilities.

15. Staff deemed the government's medium-term deficit target of 6 percent of GDP feasible by 2015, but saw implementation risks. Factors expected to support consolidation include: (i) the elimination of energy subsidies, deferred single spine wage payments, and past arrears; (ii) implementation of the envisaged tax policy measures and improvements in revenue

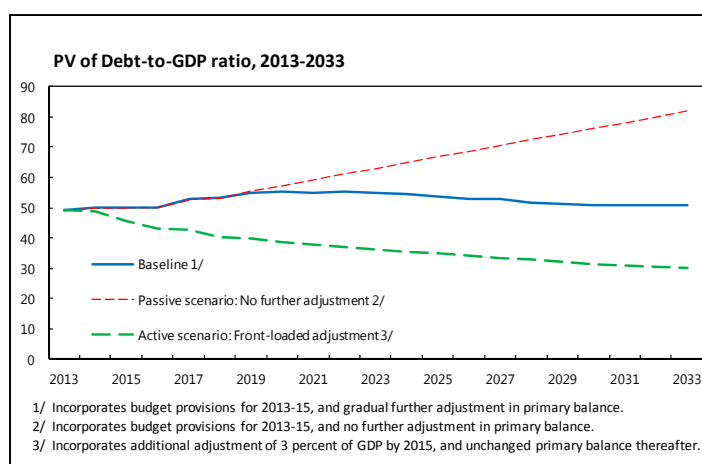
administration; (iii) full rollout of GIFMIS; (iv) complete migration of all nonsecurity subvented agencies to the mechanized payroll and set up of an integrated HR management system; (v) strengthening of cash management, with a consolidation of government accounts; and (vi) ongoing improvements in the integrated public investment and debt management framework. Success, however, depends on whether politically difficult measures—such as adjustments of energy prices to full cost-recovery levels—will be implemented in a timely and durable manner. The mission also pointed to implementation risks from possible delays and difficulties in implementing public financial management reforms, as has been the case in the past.

Projected Fiscal Savings, Change: 2012-15	
Total improvement in fiscal balance (financing basis)	5.6
Revenue and grants (positive = increase)	1.8
Revenue	2.5
Tax policy measures 1/	2.1
Tax administration	0.4
Grants (exogenous)	-0.7
Expenditure (positive = reduction)	2.6
Non-interest recurrent	0.2
Interest	-0.6
Capital	-2.0
Below-the-line changes	5.0
Deferred wage payments	2.6
Arrears clearance	2.4
Other (oil revenue, statistical discrepancy, tax refunds)	1.1

1/ Includes projected reduction of tax exemptions, and adoption of a new Value Added Tax bill, including VAT on financial services, a customs and excise tax bill, and a windfall profit tax on mining.
Source: IMF staff estimates and Ghanaian authorities.

16. Irrespective of feasibility, the mission viewed a deficit target of 6 percent of GDP as insufficiently ambitious. Informed by its external sustainability assessment, staff made the case for an additional adjustment in public consumption of 3-4 percent of GDP by 2017 to reduce the large current account deficit, while making room for increased public and private investment (Box 1). A lower fiscal and external current account deficit would also allow the buildup of official reserves toward the authorities' target of more than 4 months of imports—up from 2.8 months of imports in December 2012 (1.7 months on a net basis). This target is consistent with staff's analysis of optimal reserves, which suggests that a cover of 4.2 months of imports would provide a reasonable cushion against plausible shocks, such as deteriorating terms of trade or reversals of capital inflows invested in government bonds (Box 2).

17. An updated debt sustainability analysis supports the case for a more ambitious fiscal consolidation effort.¹ Ghana's risk of debt distress remains moderate, provided fiscal adjustment continues beyond the medium term. This would stabilize the debt ratio at its relatively high current level of about 50 percent of GDP (*baseline*). Absent further medium-term adjustment, the debt ratio would continue to rise (*passive scenario*). A more front-loaded adjustment, with additional savings of 3 percent of GDP by 2015, would set the debt ratio on a much more benign trend (*active scenario*).



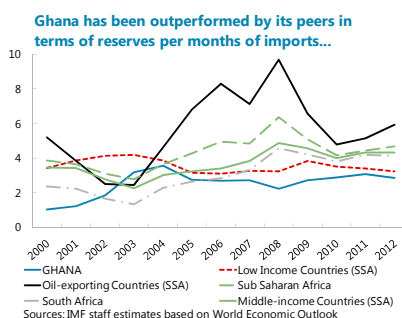
¹ See Supplement on Update of Joint IMF and World Bank Debt Sustainability Analysis.

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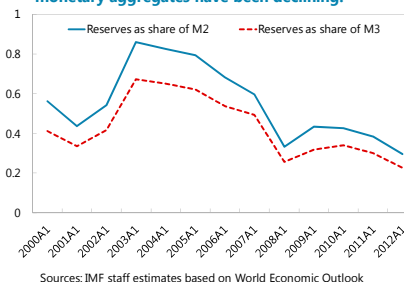
Box 1. Assessing Reserve Adequacy in Ghana

Ghana's official reserves are low compared to its sub-Saharan peer group and rule-of-thumb benchmarks.

Ghana's reserves have recovered since 2008, but still fall short of the standard import threshold of 3 months. In relation to monetary aggregates, reserves have fallen and were just above the minimum benchmark of 20 percent of M2 at the end of 2012.



...and its gross reserve holdings as a share of monetary aggregates have been declining.



While the above measures give some guidance of reserve adequacy, they have been criticized to provide only imprecise benchmarks. A conceptually superior approach is to estimate the "optimal" level of reserves which maximizes their net benefits. Following the method applied by Dabla-Norris et al. (2011), the net benefits depend on the expected cost of a crisis given the stock of reserves, a vector of fundamentals (exchange rate regime, fiscal balance, institutions), the exposure to shocks (external demand, FDI, aid), and the opportunity cost of holding reserves (interest differential).¹

Optimal reserve holdings, inputs			
Data		Shocks ¹	
Government balance, % of GDP (2012)	-11.8	External demand growth (%) ²	-1.7
CPIA, average of three last years	3.9	Terms of trade growth (%)	-13.6
Flexible exchange rate	yes	FDI as % of GDP	-4.1
Availability of IMF program	yes	Aid in % of GDP	-1.7
spread over 10-yr US treasury bond (%)	3.0		

¹ Shocks defined as one standard deviation of respective time series (2000-2012)

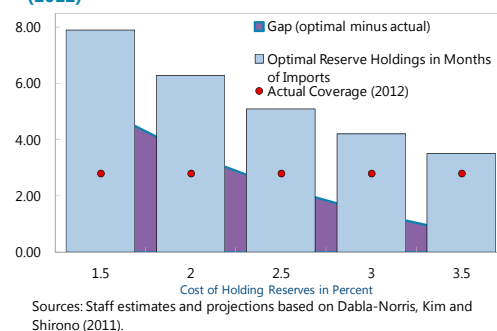
² Worst deterioration of exports since 2000 used as high s.d. driven by large upward swings

Source: IMF staff estimates based on Dabla-Norris, Kim, and Shirono (2011).

The approach implies optimal reserve holdings of 4.2 months of imports (\$8.1 billion) for 2012, compared with actual reserves of 2.8 months of imports (\$5.3 billion). Raising reserves to the suggested level would reduce the predicted cost of a crisis, in the event of a large shock, by about 24 percent.

The assessment of Ghana's reserve buffer must also be seen in the context of strong terms of trade in 2012, increased foreign holdings of government paper, and short-term central bank liabilities. Oil, gold, and cocoa account for 85 percent of Ghana's total exports of goods, and the value of gold and cocoa exports alone has risen by more than 90 percent since 2009. A return of gold and cocoa exports to their 2009 levels could curb net current account inflows by \$1.6 billion. At the same time, high yields on government bonds have increased foreign participation, with nonresidents now holding about one-third of government domestic debt (in three and five-year bonds). This creates potential for net portfolio inflows, as bonds are typically held to maturity and those maturing in 2013 had very low foreign participation. However, if instead foreigners decided to reduce their exposure by selling, for example, half of their holdings (arguably at a significant discount) on the secondary market, this would translate into capital outflows of \$1.4 billion. Finally, gross reserves are currently encumbered by the need to rollover short-term swaps and bridging facilities, equivalent to \$0.8 billion, with \$0.5 billion maturing in 2013. Thus, a confluence of adverse developments could seriously weaken the balance of payments, lending strong support to the model-based conclusion that current reserve levels provide an inadequate buffer against plausible country-specific shocks.

Ghana: Optimal Reserve Holdings in Months of Imports (2012)



¹ See Dabla-Norris, Kim and Shirono (2011): "Optimal Precautionary Reserves for Low-Income Countries: A Cost-Benefit Analysis", WP/11/249.

Box 2. External Sustainability Assessment¹

Medium-term current account benchmarks for Ghana range from -4 to -2.8 percent of GDP. Ghana's optimal path in the next five years should lead to a reduction in the current account deficit combined with a significant fiscal consolidation and a shift from public consumption to public investment.

The three methodologies used—external sustainability (ES), macroeconomic balance (MB), and model-based—approach the assessment from different perspectives. Combining their results provides a richer picture of Ghana's external position. The ES approach provides a current account benchmark consistent with stabilizing NFA at a certain level, but abstracts from the country's underlying internal fundamentals. The MB approach estimates a benchmark based on the country's fundamentals relative to its trading partners, but does not capture Ghana's recent oil windfall and its potential use for infrastructure investment. In a capital-scarce economy, like Ghana, this may warrant higher current account deficits. By calibrating the dynamic neoclassical model by Araujo et al. (2012) to the case of Ghana, the recent discovery of offshore oil and gas reserves is explicitly incorporated into the assessment of the current account.

Both the ES and MB approaches suggest that the current account deficit should decline in the medium run, but magnitudes of adjustment differ. Stabilizing Ghana's NFA-to-GDP ratio at its 2009 pre-oil discovery level of -26.8 percent implies a current account benchmark of -2.8 percent of GDP. The MB approach predicts a medium-run benchmark of -3.5 percent of GDP which is strongly influenced by Ghana's relatively high rates of GDP per capita and population growth. Under this approach a fiscal adjustment would narrow the gap from two sides: directly, by reducing the actual current account deficit, and indirectly by lowering the benchmark, with every percentage point improvement in Ghana's fiscal balance translating into a 0.3 percentage point increase in the sustainable current account deficit.

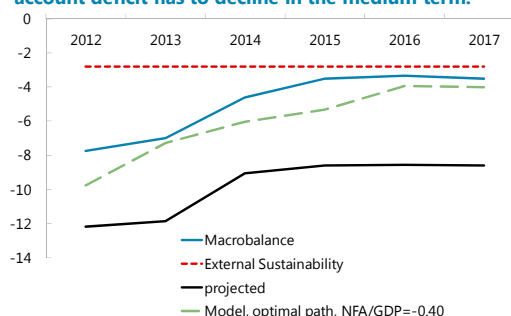
A model-based approach is used to qualify the above results and to provide quantitative indications of the optimal composition of consumption and investment in Ghana in the medium run.² The paths of the current account, as well as private and public consumption and investment implied by the model, are the result of the optimal choice of a social planner maximizing social welfare. The optimal choice takes account of the need for infrastructure scaling up and incorporates inefficiencies inherent in investment projects in developing countries, given a long-run level of NFA (note that FDI is treated as decreasing NFA in the model). Once calibrated to Ghana's economy, the model can provide a current account benchmark, as well as information on the optimal composition of consumption and investment.

First, the optimal composition of spending is estimated consistent with the projected current account deficit in the medium run. This step reveals the long-run level of NFA under which the currently projected deficit for 2017 would be optimal and the optimal levels of private and public consumption and investment given this projection. A current account benchmark broadly in line with the projected path of the current account deficit in the medium run is consistent with a long-run NFA level of -78 percent of GDP. This level would place Ghana in the lowest 20th-percentile within the group of middle income countries.³ Given this long-run NFA level, the model predicts that recurrent government expenditure is around 4.5 percentage points of GDP above and public investment more than 2 percentage points below the optimal value in 2012. In 2017, government consumption is projected to be more than 3 percentage points of GDP higher than its benchmark and public investment almost 0.5 percentage point below its optimal value.

	MB	ES	Model
short-run (2012)			
CA projection	-12.2%	-12.2%	-12.2%
CA benchmark	-7.7%	-2.8%	-9.8%
CA gap	-4.4%	-9.4%	-2.4%
medium-run (2017)			
CA projection	-8.6%	-8.6%	-8.6%
CA benchmark	-3.5%	-2.8%	-4.0%
CA gap	-5.1%	-5.8%	-4.6%

Sources: Staff projections and calibration based on Araujo et al. (2012).

Current Account: All approaches suggest that the current account deficit has to decline in the medium term.



Sources: Staff projections and calibration based on Araujo et al. (2012).

Box 2. External Sustainability Assessment (Concluded)

Second, the spending composition and the current account benchmark are derived based on a more conservative long-run NFA to GDP ratio. The optimal macroeconomic path is now derived assuming an NFA-to-GDP ratio of -40 percent corresponding to the median NFA level of lower-middle income countries. This NFA level implies a benchmark for the current account deficit of 9.8 percent of GDP in the short run and 4 percent of GDP in the medium run.

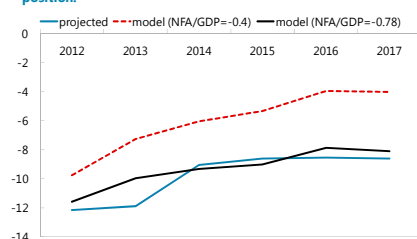
The medium-term adjustment of the current account occurs mainly through an optimal decline in public and private consumption to 8.8 and 75.6 percent of GDP in 2017. It implies that public consumption would have to adjust by 3.5 percentage points more than currently projected, whereas public investment would be at its optimal level in 2017. While the model predicts consumption shares to fall on an optimal path, the decline would have to be much stronger without the discovery of oil, i.e. the effect of the oil windfall on private and public consumption is positive (about 4 percent of GDP for private and 0.5 percent of GDP for public consumption in the medium term). The model also suggests room to increase private investment by 0.5 percentage point of GDP in both the short and medium run.

Medium-Term Current Account Adjustment Model ¹ vs. Projections (In percent of GDP)				
	Projected	Model	Adjustment	
Current account	-8.6	-4.0	4.6	
Public consumption	12.3	8.8	-3.5	
Public investment	8.9	9.0	0.1	
Private consumption	78.4	75.6	-2.7	
Private investment	12.0	12.5	0.5	

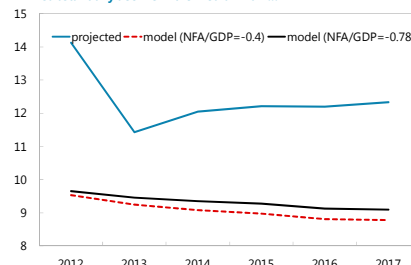
¹ NFA/GDP=-0.40

Sources: Staff projections and calibration based on Araujo et al. (2012)

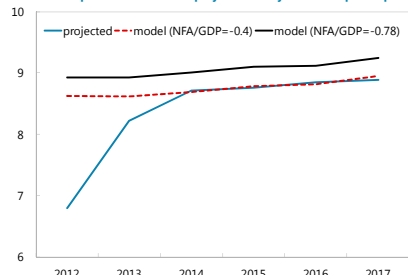
Current Account (in Percent of GDP)
The current account has to improve if Ghana targets to be at least not worse than half of lower middle income countries in its NFA position.



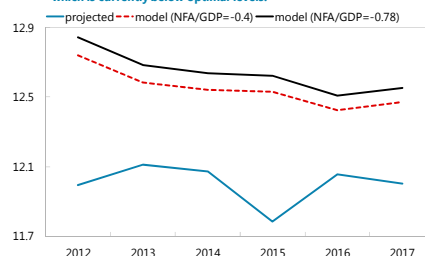
Government Consumption (in Percent of GDP)
Non-investment expenditures have to substantially decline in the medium run ...



Government Investment (in Percent of GDP)
... while public investment is projected to adjust to the optimal path.



Private Investment (in Percent of GDP)
There is room to further improve private investment which is currently below optimal levels.



Sources: IMF staff projections and calibration based on Araujo et al. (2012)

The current account deviations from the benchmarks obtained above suggest that the Ghana cedi is overvalued by 13.6 to 17.1 percent in the medium run, but call for different policies to support external sustainability. In particular, both MB and model-based approaches suggest that the adjustment in the current account can be achieved through fiscal consolidation, with the latter approach highlighting that the adjustment of fiscal expenditure should arise through a disproportionate cut in current expenditure, while public and private investment should be safeguarded.

¹ Based on research by Bin Grace Li, Monique Newiak, Geneviève Verdier, and Felipe Zanna.

² The quantitative results depend partly on the specific assumptions made about the efficiency of public and private investment, as well as the calibration of the adjustment cost. The possible effects of these parameter changes are explored in Araujo et al. (2012).

³ Maria Milesi-Ferretti (2007): The External Wealth of Nations Mark II. Revised and Extended Estimates of Foreign Assets and Liabilities, 1970-2004, Journal of International Economics, 73, 223-250.

18. A model-based analysis confirms these conclusions and provides further insights on the trade-offs between different policy choices.

Appendix III presents the results of a dynamic stochastic general equilibrium (DSGE) model, capturing the interaction between public investment, growth, and debt sustainability, calibrated to the Ghanaian economy. The analysis illustrates that achieving a long-term debt ratio of about 50 percent may require a stronger fiscal effort than assumed in the standard DSA. It confirms that further upfront consolidation would reduce the debt ratio significantly with only small reductions in the growth dividend. Alternative policy scenarios suggest that such a front-loaded adjustment could generate significant benefits if combined with reforms to enhance investment efficiency. The latter would even create scope for a more ambitious scaling up of public investment and a larger growth dividend, without endangering the debt dynamics. A more ambitious investment profile without improved efficiency, however, would raise the debt ratio significantly.

19. In light of these findings, the mission recommended policy measures to generate savings of another 3 percent of GDP by 2015, complemented by further institutional reforms.

While there are limits to achieving faster consolidation in 2013, staff viewed prompt action on measures that can be implemented quickly as critical to reduce fiscal and external risks. This, together with an ambitious and credible action plan for the coming years, would go a long way in reinforcing confidence. The mission suggested a thorough expenditure review, with possible support from the Fund's Fiscal Affairs Department, to identify areas for savings and reprioritization, consistent with the government's own agenda.

20. There was broad consensus on the areas for reform:

- **On the expenditure side**, it was agreed that a return to *full cost-recovery pricing for fuel and utilities*—cushioned by targeted social programs—should be pursued as soon as possible, not only to generate savings, but also to restore the financial viability of the public energy companies, as part of a comprehensive reform of the sector. The largest potential for savings could be achieved by *rationalizing the wage bill*, including actions to streamline, discontinue, or commercialize the activities of subvented agencies. A public sector reform secretariat has been established under the office of the president to develop specific plans. The authorities saw benefits in the mission's proposal to move to multi-year wage agreements and to negotiate salary increases before the budget is finalized to enhance predictability. Decisive action on these and other fronts, in the context of a broad public expenditure review, could reasonably generate savings of another 2 percent of GDP by 2015. On wages alone, savings of 1¾ percent of GDP could be achieved, relative to current projections, by limiting pay increases to inflation, while keeping the size of the public sector unchanged.
- **On the revenue side**, a far-reaching removal of exemptions, together with further tax policy measures, for example in the area of property taxes, could generate additional revenue of 1 percent of GDP by 2015.

These measures, taken together, would help rebuild external buffers, establish favorable debt dynamics, improve the composition of fiscal spending toward priority areas, and allow a readjustment in the policy mix to reduce high real interest rates.

GHANA

21. The authorities also recognized the importance of institutional reforms to strengthen budgetary control and predictability:

- **Debt management and investment planning.** Building on past achievements, the authorities are committed to further strengthen their performance in these key areas that will only increase in importance with growing reliance on nonconcessional financing. The planned issuance of another Eurobond of up to \$1 billion is part of their strategy to smooth rollover needs and reduce debt-servicing costs—by replacing more expensive borrowing from the less liquid domestic market—while shoring up international reserves. The authorities are also taking a more systematic approach to the implementation of investment projects, prioritizing completion of existing over new projects.
- **Budget allocation rules for statutory funds.** Fixed allocation to these funds, established for priority areas, such as roads, education, and fiscal decentralization, limit budget flexibility but are constitutionally mandated. Rather than attempting to change the underlying principle, the authorities saw scope in reviewing the transfers and associated spending obligations, with a view to streamlining overall government expenditure and creating more flexibility to adjust spending in response to shocks or changed priorities.
- **Fiscal rules.** The authorities were open to renew the dialogue over the adoption of comprehensive fiscal rules to support fiscal prudence and debt sustainability, once consolidation is firmly on track and fiscal predictability has been established.

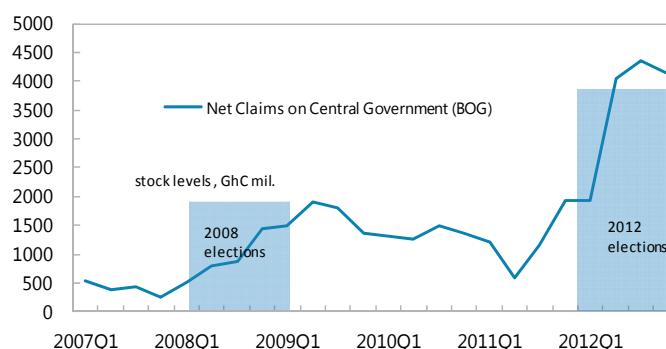
MONETARY POLICY: LITTLE ROOM FOR EASING

There was broad agreement that monetary policy needs to remain tight until fiscal consolidation is firmly established.

22. The mission raised concerns about the BoG's direct financing of the fiscal deficit in the run-up to the 2012 elections.

The sharp increase in net credit to the government far exceeded the statutory limit of 10 percent of revenue for *total* bank financing and risked weakening the perception of the BoG's independence, as well as the credibility of the inflation targeting regime.

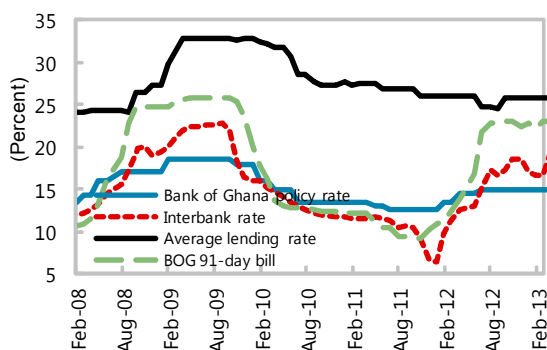
The authorities explained that the high recourse to BoG financing reflected revenue shortfalls at the end of 2012, when the BoG had already covered government payments on the basis of higher revenue projections. Going forward, they expected that ongoing improvement in cash management and forecasting would allow them to



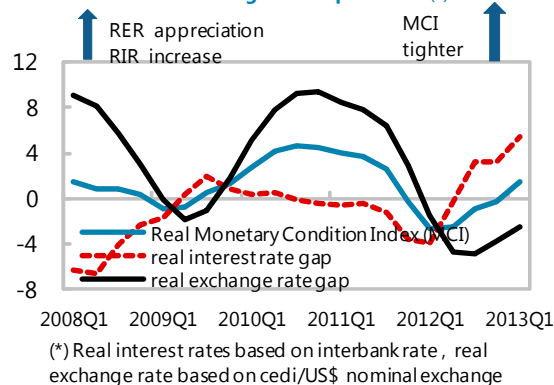
Sources: Bank of Ghana and IMF staff estimates.

Figure 4. Ghana: Monetary and Financial Indicators

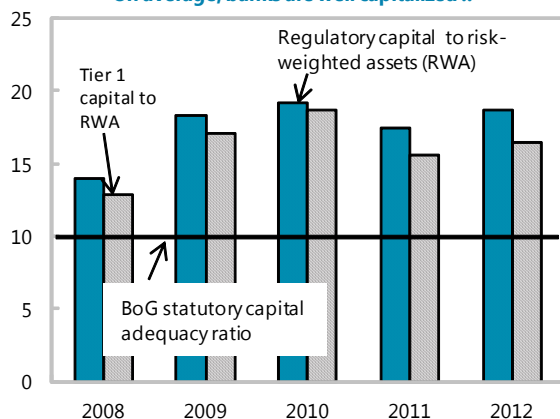
In the second half of 2012, short-term market rates rose sharply...



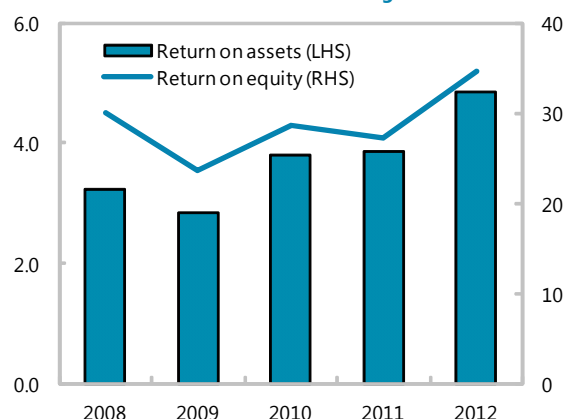
...and monetary conditions tightened, with higher real interest rates and the reversal of the earlier real exchange rate depreciation(*).



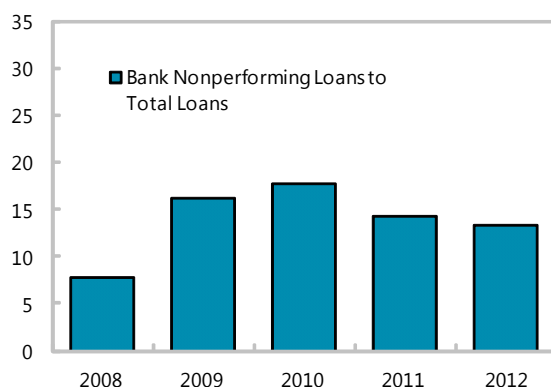
On average, banks are well capitalized ..



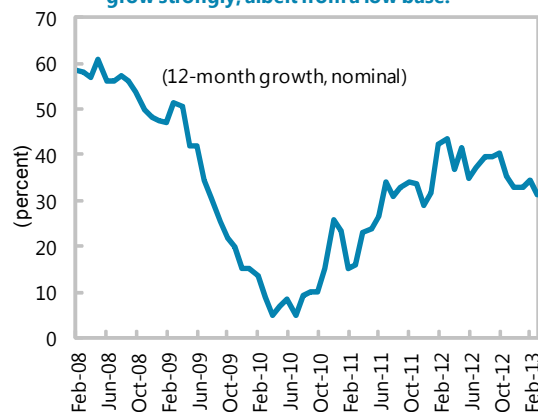
...and returns are strong.



NPLs declined but remain high...



...while private sector credit has continued to grow strongly, albeit from a low base.



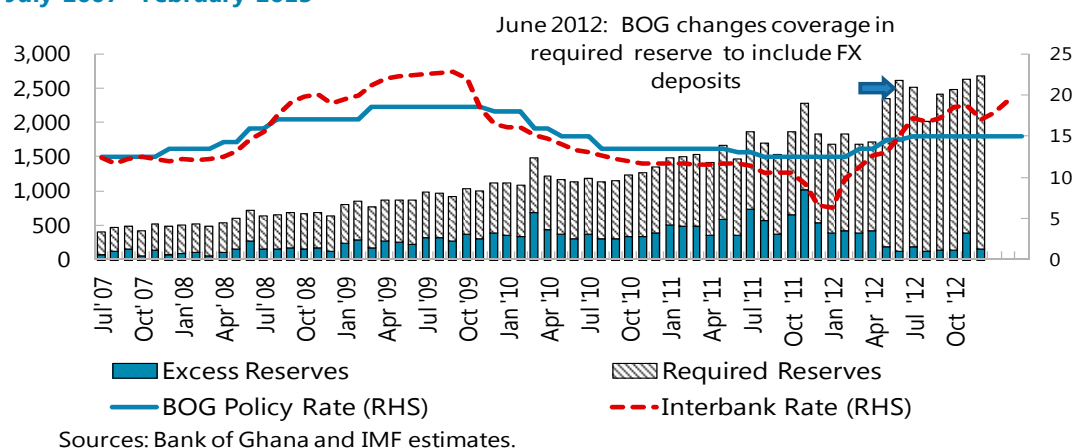
Source: Ghanaian authorities; DataStream; and IMF staff estimates.

GHANA

significantly reduce such unintentional recourse to BoG's financing, but did not think that a full elimination was practical. The mission agreed with the BoG's proposal to establish a separate ceiling for its component of government financing that is consistent with macroeconomic stability objectives and enforced.

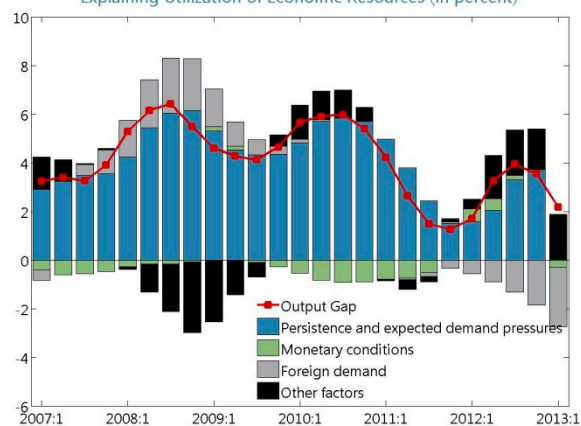
23. Faced with a rapidly depreciating currency, the BoG took a number of measures in 2012 to tighten domestic liquidity. Apart from raising its policy rate by 150 basis points and directly mopping up liquidity, the BoG reduced banks' maximum net open forex position and required them to hold mandatory reserves on foreign currency deposits in cedi, rather than foreign currency. These measures were successful in absorbing excess liquidity and stabilizing the cedi, while raising interbank rates well above the policy rate. Thus, the effective policy stance, as evidenced by market rates, has been tighter than suggested by the policy rate.

Ghana: Bank Liquidity and Interest Rates
July 2007 - February 2013



24. There was agreement that interest rates will need to remain high until inflationary expectations decline and fiscal consolidation is firmly established. Both actual inflation and inflation expectations have risen recently in the context of aggregate demand pressures from expansionary fiscal policies, consistent with staff's assessment of a positive output gap (text chart and analysis in Box 3). With upside risks from the sharp increase in reserve money, driven by net credit to the government, the mission advised keeping a tight monetary stance at this juncture.

Figure 6. Ghana Output Gap and Factors
Explaining Utilization of Economic Resources (in percent)



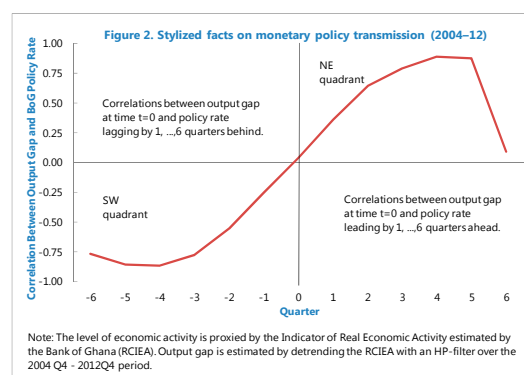
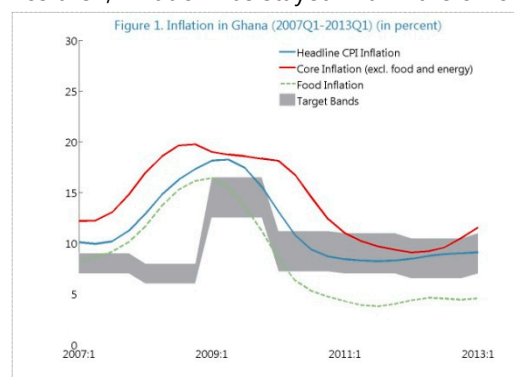
Box 3. Six Years of Inflation Targeting in Ghana: Policy Lessons and Challenges¹

Ghana is one of the few lower-middle income countries that adopted inflation targeting (IT) as a formal framework for its monetary policy. This box summarizes the key findings of a joint study by the IMF African and Research Departments on the performance of Ghana's IT regime over the six years of its existence (2007–12). It applies a stylized macroeconomic model of the kind typically used by central banks to guide their policy in the early phases of their IT experience.² The key findings and policy conclusions are: the IT regime has been successful in bringing inflation down to single digits, but there remain significant operational challenges to make the framework more transparent and accountable.

Inflation Record and Monetary Transmission

The IT regime has been successful in delivering on its goal of reducing inflation and keeping it stable at the authorities' target level. The Bank of Ghana (BoG) reduced headline inflation from 20-percent levels in 2004 to 10.7 percent by end-2010 (above the 8.6 percent mid-point target, but within the target band); since then, inflation has stayed within the official target bands (Figure 1). This was helped, however, by low food inflation and administered energy prices; core inflation, excluding food and energy, has been persistently in double digits throughout the IT period (2007-to date) and within the upper limit of the official target bands since 2011.

The interest rate channel, a key channel of IT policy, seems to function in a standard way. This conclusion takes account of the fact that the overnight interbank market rate has at times deviated quite significantly from the official policy rate. Because the former is a better proxy of the BoG's effective policy stance, it is used instead of the official policy rate to analyze the implementation of the IT framework. Figure 2 shows the correlation between the interbank market rate and economic activity, measured by the output gap (the difference between actual output and its potential level). Past interest rate tightening is associated with moderating economic activity in the future (Figure 2, SW quadrant showing negative correlation), indicating that the BoG is forward-looking by tightening rates when an output gap is expected to open up. Also, the BoG seems to appropriately tighten the policy rate in response to past increases in the output gap (Figure 2, NE quadrant showing positive correlation).



Source: Bank of Ghana and IMF Staff estimates

Interpreting interest rates, inflation, and economic activity over the IT period

The six-year experience of IT in Ghana can be divided into three sub-periods: (i) transition period (2007–2008), (ii) IT-gaining-credibility period (2009–2010), and (iii) complacent period (2011–2012). A macroeconomic model is used to shed light on the key drivers of economic cycles and the policy transmission mechanism during those periods. The model reflects the essence of the macroeconomic relationship between output, prices, interest rates, and exchange rates as predicted by economic theory. Specifically, monetary policy follows a Taylor-type rule in which the policy interest rate responds to expected future deviations of inflation from the target and to the output gap. Formally, the behavior of output and inflation is captured by an IS curve, which captures demand shocks, including fiscal policy changes, and a Phillips curve. As a small open economy, Ghanaian economic activity is further affected by foreign factors. The exchange rate, satisfying risk-adjusted uncovered interest-rate parity, is a nominal shock absorber between the domestic and foreign developments.

Box 3. Six Years of Inflation Targeting in Ghana: Policy Lessons and Challenges (Concluded)

The model can explain volatility in interest rates, prices, and output by underlying (model-estimated) “fundamentals.”

Figure 3 shows the actual interbank rate and the fundamental rate estimated by the model. Figure 4 decomposes the interbank (policy) rate into its structural components, including unexpected shocks (other factors). During the IT transition period, the actual rate was close to its fundamental level, but policy still lacked credibility: when the food and fuel price shock hit the economy in 2008, inflation expectations surged (Figure 5). This prompted the authorities to increase rates by more than warranted by economic fundamentals (Figure 3) in an attempt to gain credibility and anchor inflation expectations. During the following five quarters (2009Q3–2010Q4), inflation declined, reaching its 9 percent target by end-2010. Subsequently, the monetary stance turned too accommodative, planting the seeds for yet another credibility test. As the economy entered an expansionary period with the start of oil production, liquidity conditions eased rapidly. The interest rate fell below its fundamental rate, and associated capital outflows triggered a sharp exchange rate depreciation. Inflation expectations accelerated again (light green bars in Figure 5). At this point, breaking the cycle required aggressive action, leading to double-digits real interest rates.

What are the conclusions?

Stronger commitment to single-digit inflation could have benefited the real economy through lower financing costs. The ‘lack of credibility’ premium in interest rates, comes from different sources: an imperfect alignment of market rates with the BoG policy rate, delayed reaction to growing liquidity pressures, and insufficient communication of the policy stance. Credibility of the IT framework could also be enhanced by setting a medium-term inflation target to better anchor expectations. The BoG may want to adopt a more ambitious medium-term inflation target of 5-6 percent, to realize the significant welfare gains from a lower inflation environment. Looking forward, the BoG should make further improvements in the systematic use of a model-based forecasting and policy analysis (FPAS) and in communicating effectively to the public its policy decisions, including policy and forecasting mistakes, in line with best practice in IT-based monetary policy frameworks.

Considerations beyond the model-based analysis

A successful implementation of the IT framework cannot be seen in isolation from fiscal policy shocks which may threaten the inflation target. An expansionary fiscal policy that relies directly on BoG financing, as it has occurred in Ghana in the past two elections (2008 and 2012), makes it difficult to achieve and sustain lower inflation rates. Such a policy of monetizing the fiscal deficit raises inflation directly, while also entrenching it through higher inflationary expectations.

Figure 3. Actual Interbank Rate vs Model-based “Fundamental” Rate (in percent)

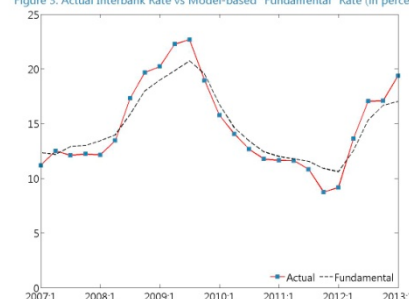


Figure 4. Factors Explaining Interest Rate Changes (in percent)

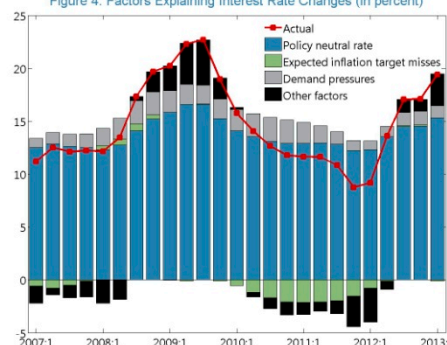
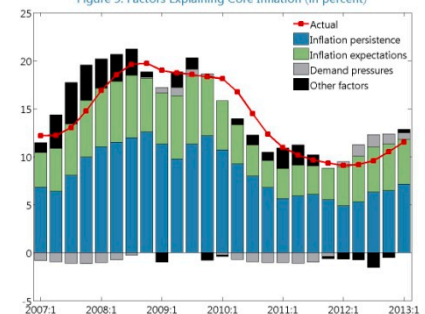


Figure 5. Factors Explaining Core Inflation (in percent)



Source: Bank of Ghana and IMF Staff estimates.

¹The authors of this box are Alfredo Baldini (AFR) and Martin Fukac (RES).

²See Douglas Laxton, Alasdair Scott, and David Rose, 2009. “Developing a Structured Forecasting and Policy Analysis System to Support Inflation-Targeting (IFT),” IMF Working Papers 09/65, International Monetary Fund.

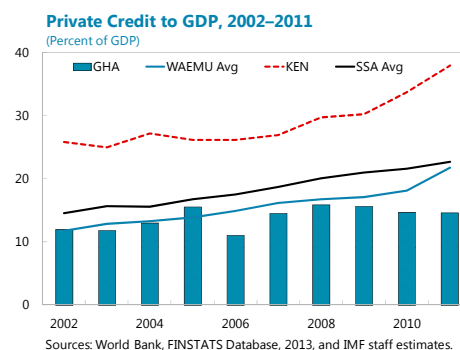
It recommended an increase in the policy rate to counter inflationary pressures, supported by active liquidity management that keeps market rates appropriately tight. This would also strengthen the signaling role of the policy rate within the inflation-targeting framework by narrowing the gap with current interbank rates. The mission suggested that a further tightening of the policy stance could be warranted, should recent hikes in domestic fuel prices create second-round effects, or should the cedi come under renewed pressure. In the latter case, the mission cautioned against direct interventions that would further deplete an already low reserve cover. Subsequent to the mission, during its May meeting, the Monetary Policy Committee (MPC) raised the policy rate by 100 basis points to 16 percent.

25. The mission also advised the BoG to strengthen the implementation of the inflation-targeting framework. It welcomed the BoG's interest in improving its liquidity management, based on a consistent and user-friendly liquidity-forecasting framework. To strengthen its policy effectiveness, the BoG should manage liquidity in a manner that results in a close alignment of market rates with its policy rate. Following the May MPC meeting, the BoG announced two welcome changes to its monetary operations to strengthen the effectiveness and signaling functioning of the policy rate: (i) a widening of the interest rate corridor by raising the upper bound to 200 basis points above the policy rate (with the lower bound remaining at 100 basis points); and (ii) introduction of a standing facility to manage liquidity more effectively in the interbank market, enhancing the transmission mechanism. Looking forward, the mission encouraged the BoG to make further improvements in the systematic use of a structural model-based inflation forecasting tool, and in communicating effectively to the public its policy decisions—including policy and forecasting mistakes—in line with best practice in IT-based monetary policy frameworks. Finally, it suggested that credibility of the IT framework would be further enhanced by setting a one- to two-year rolling inflation target, as opposed to the current end-year target, to better anchor expectations. The mission also noted that a more ambitious target could generate significant welfare gains from a lower inflation environment.

FINANCIAL SECTOR: CONTAINING VULNERABILITIES AND REMOVING STRUCTURAL CONSTRAINTS

The Bank of Ghana is focused on strengthening its regulatory and supervisory powers, in line with the 2011 FSAP recommendations, and to keep pace with new risks in the financial sector. The mission's advice benefited from enhanced financial sector surveillance (Box 4).

26. The authorities shared the mission's concerns about high interest rates pricing out profitable investment projects. While private credit has grown strongly in 2012, this has occurred from a low base. Banks' lucrative investments in government securities have reduced their incentives to actively seek lending



GHANA

opportunities to the private sector. The authorities fully agreed that decisive fiscal consolidation was necessary to reduce the crowding out of private investment.

27. While the banking system has grown steadily in assets and profitability, the mission cautioned about the buildup of new risks. Despite strong credit growth, the ratio of non-performing loans (NPLs) has not fallen since mid-2012, but merely stabilized at around 13 percent, and high interest rates spell risks of a renewed deterioration in banks' credit portfolios. Moreover, strong profits from high treasury-bill rates can reduce banks' incentives to improve efficiency,

Box 4. Enhanced Financial Sector Surveillance: Main Conclusions¹

Ghana was selected as one of the pilot cases for enhancing surveillance of financial systems in low-income countries. The purpose is to go beyond the traditional surveillance focus on banking system soundness and solvency, by analyzing in more depth the interplay between financial development, macroeconomic and financial stability, and effectiveness of macroeconomic policies. The main conclusions of the study are as follows:

- While the banking system has grown rapidly and is competitive by standard metrics (barriers to entry and concentration of assets), private credit has remained relatively low as a share of GDP and access to affordable credit is a major constraint on growth.
- High real interest rates—a result of structural factors, such as high operating cost, but also excessive government borrowing—are constraining private sector access and create risks of a renewed increase in nonperforming loans (NPLs).
- An upgrading of financial sector legislation and supervisory practices is ongoing and is needed to deal with the growing complexities of an evolving financial landscape with increased foreign participation and a growing role of microfinance institutions.
- Rural and Community Banks have become a main channel for financial inclusion, though only about 30 percent of adults have an account at a formal institution. A sizeable part of the population relies instead on the services of about 600 microfinance companies, as well as 3,000-5,000 individual susu collectors that serve over half a million customers.
- The ownership role of the state, especially of the BoG, in the financial sector raises concerns about a level playing field and reputational risks to the BoG's credibility as an independent regulator and supervisor. Risks also arise from the slow decision-making and resolution process for the two remaining weak banks.
- The BoG has the tools to conduct effective inflation targeting, but there is room to improve its forecasting and liquidity management framework to restore the signaling role of its policy rate.
- A return to fiscal discipline and effective policy coordination are needed for successful further disinflation and increased financial deepening.

¹ See accompanying Supplement on Enhancing Financial Sector Surveillance.

potentially delaying otherwise warranted consolidation and cost reduction in the industry. Overhead costs relative to total assets are already significantly above peer countries, and the cost-to-income ratio is high. While, on average, only 30 percent of banks' income results from treasury bills, retaining profitability in some institutions may prove challenging, once the government's high borrowing demands are reversed. Concentration is also high with large exposures to single obligors and economic sectors. In light of these risks, the BoG is encouraging banks to make effective use of the existing credit reference bureaus and was open to suggestions to raise minimum capital buffers against a future downturn.

28. Building on recent improvements in supervisory practices, the mission urged the BoG to focus on the remaining gaps and challenges identified in the 2011 FSAP.² The BoG has greatly improved its offsite supervision, backed by enhanced risk-assessment processes. It is now focusing on strengthening its regulatory powers and standards in areas such as consolidated supervision and has requested Fund technical assistance to help address gaps and inconsistencies in the laws that constrain its ability to set and enforce prudential regulations. Further areas for improvement include deepened cross-border supervision, where the BoG has signed MoUs and begun to supervise some foreign-owned banks jointly with its relevant supervisory counterparts. Finally, while the number of weak banks has declined since the 2011 FSAP, the mission encouraged the authorities to address the risks posed by the two remaining problem banks, either through their sale or orderly liquidation, while developing contingency plans in case of further stress. This would also reduce reputational risks to the BoG's credibility as a supervisor, arising from its ownership stake in one of the banks.

STAFF APPRAISAL

29. Despite a strong growth momentum, short-term stability risks are significant. Ghana's strong democratic credentials and favorable prospects for oil and gas production continue to attract significant FDI. Yet, low external buffers and a rising domestic debt ratio expose the economy to risks, such as weaker terms of trade, reduced capital inflows, or unanticipated spending needs. Energy sector problems could curtail growth, while excessive domestic government borrowing is raising the cost of credit to the private sector. Both factors have been identified as key constraints to growth.

30. Ghana's positive medium-term outlook is contingent on strong political will to confront these challenges decisively. The immediate policy priority is to safeguard stability by rebuilding fiscal and external buffers. Decisive fiscal consolidation will, in due course, also allow for a reduction in interest rates. Going forward, successful economic transformation will require a realignment of spending away from wages and subsidies toward investment in infrastructure, while structural fiscal reforms are needed to restore policy credibility and build institutional resilience to the political cycle. Ultimately, better infrastructure, improved access to credit, further investment in

² See Appendix IV on the status of implementation of key FSAP recommendations.

GHANA

health and education, and sustained macroeconomic stability will be central for Ghana's ambition of achieving full middle-income status and raising the living standards of all its citizens.

31. The government's primary policy task is to reverse the large fiscal slippages of 2012.

Gaining control over the wage bill is the first priority, starting with a thorough examination, already initiated by the government, of the factors that drove the spending overrun in 2012. Improvements in revenue administration and new tax measures are welcome, but much of their impact is likely to be felt only in the coming years. The announced reintroduction of the stabilization levy combined with other immediate measures will provide an important offset for early-year revenue shortfalls, but prompt action is also needed to curtail tax exemptions and bring fuel and utility prices back to full cost-recovery levels. The latter is essential not only to eliminate costly subsidies, but also to restore the financial viability of the power sector companies as a requirement for tackling Ghana's energy problems. In the absence of additional policy measures, and assuming a delayed removal of energy subsidies, staff sees significant risks that this year's deficit target of 9 percent of GDP will be missed.

32. The government's deficit target of 6 percent of GDP by 2015 is achievable, but would keep public debt high and external buffers low. Staff recommends an additional fiscal adjustment of 3 percent of GDP by 2015, using a combination of revenue and expenditure measures, including a reduction in the wage bill as a share of GDP to make room for investment in critical infrastructure and social priority areas. Without the additional fiscal adjustment, Ghana's public debt burden would likely remain high, and official reserves would continue to fall short of the authorities' target of 4 months of imports, leaving the economy exposed to terms-of-trade shocks and destabilizing shifts in confidence.

33. Staff's external stability assessment confirms the need for reducing recurrent government spending. Ghana's projected medium-term current account deficit exceeds the sustainability benchmarks, suggesting an overvaluation of the real exchange rate by 13½ to 17 percent in the absence of further policy adjustment. An additional reduction in public consumption of 3-4 percent of GDP by 2017, would facilitate a reduction in the current account deficit to a sustainable level of about 4 percent of GDP, without the need for an exchange rate adjustment, while making room for increased public and private investment.

34. The monetary policy stance will need to remain tight until inflationary pressures decline and fiscal consolidation is firmly established. Both actual inflation and inflation expectations have risen recently, with upside risks from the sharp increase in government borrowing and a still positive output gap. The latest increase in the policy rate was warranted to counter inflationary pressures and should continue to be supported by active liquidity management to keep market rates appropriately tight. A further tightening of the policy stance could be warranted in the event of second-round effects from recent hikes in domestic fuel prices or renewed pressure on the cedi, while foreign reserves should be protected. Successful fiscal consolidation would provide scope for lowering interest rates in due course.

35. The Bank of Ghana has room to improve the functioning of its inflation targeting framework. Recent decisions to raise the policy rate, combined with measures to steer market rates back into the policy corridor, are welcome and important steps to restore the policy rate's signaling role within the inflation-targeting framework. Going forward, the BoG should manage liquidity consistently in a manner that maintains a close alignment of market rates with the policy rate, with the latter being set at a level consistent with the inflation target. The process for selecting the policy rate could be strengthened by the systematic use of a model-based inflation forecasting tool which—together with improved communication to the public—would bring the BoG closer to best practice in IT-based monetary policy frameworks. Credibility of the IT framework could be further enhanced by setting a one- to two-year inflation target to better anchor expectations, while a more ambitious target could generate significant welfare gains from a lower inflation environment.

36. The priority in the financial sector is to contain vulnerabilities and remove structural constraints. Ghana's banking sector has grown steadily in assets and profitability, but high interest rates are pricing out profitable investment projects, while raising the risk of a renewed increase in nonperforming loans. Higher minimum capital buffers may be warranted as protection against a future downturn. The BoG has implemented important actions over the past years to improve bank supervision. It should now focus on addressing gaps and inconsistencies in the banking laws, and deepening cooperation with regional counterparts to improve the regulation and supervision of foreign banks active in Ghana. Dealing with the remaining two weak banks is also a priority, as is the divestiture of the BoG's financial stake in banks that it supervises.

37. It is recommended that the next Article IV Consultation be held on the regular twelve-month cycle.

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Table 1. Ghana: Selected Economic and Financial Indicators, 2011–18

	2011	2012		2013	2014	2015	2016	2017	2018
		6-7 ECF	Est.			Proj.			
(Annual percentage change; unless otherwise indicated)									
National account and prices									
GDP at constant prices 1/	15.0	8.2	7.9	7.9	6.1	5.5	5.1	5.1	8.0
Real GDP (nonoil)	9.4	8.8	7.8	5.9	6.0	5.9	5.8	5.8	5.8
Real GDP per capita	12.1	4.5	5.2	5.2	3.4	2.9	2.5	2.4	5.3
GDP deflator	13.0	15.3	13.3	13.7	12.8	11.1	10.3	8.8	7.4
Consumer prices									
Consumer price index (annual average)	8.7	9.8	9.2	10.3	9.8	8.2	7.5	7.2	7.0
Consumer price index (end of period)	8.6	11.5	8.8	10.8	8.8	7.7	7.3	7.1	6.9
Consumer price index (excl. food, annual average)	15.0	14.3	11.6	11.0	7.8	8.2	8.0	7.6	7.3
Money and credit									
Net domestic assets ²	28.7	29.1	51.9	44.8	19.5	15.6	17.6	17.4	12.1
Credit to the private sector ²	29.0	20.1	32.9	38.4	21.7	28.0	16.7	9.0	19.2
Broad money (M3, including foreign currency deposits)	32.2	34.8	24.3	27.8	23.4	20.4	19.7	19.1	18.5
Velocity (GDP/M2, end of period)	3.1	3.7	3.0	2.9	2.8	2.7	2.7	2.6	2.5
Base money	31.1	25.1	36.0	24.0	21.7	18.4	17.1	16.3	15.3
Banks' lending rate (weighted average; percent)	25.9	...	25.7	...	...	...	...	...	...
Policy rate (in percent, end of period)	12.5	...	15.0	...	...	...	...	...	...
(Percent of GDP)									
National accounts									
Gross capital formation	18.5	19.8	18.8	20.3	20.8	20.6	20.9	20.9	21.3
Government	6.2	7.8	6.8	8.2	8.7	8.8	8.9	8.9	9.3
Private	12.3	12.0	12.0	12.1	12.1	11.8	12.1	12.0	12.0
National savings	9.4	10.7	6.6	8.5	11.7	11.9	12.3	12.3	13.9
Government	5.5	5.6	0.3	5.2	4.9	5.1	5.3	5.4	5.8
Private ³	3.8	5.1	6.3	3.2	6.8	6.8	7.0	6.9	8.0
Foreign savings	-9.1	-9.1	-12.2	-11.9	-9.1	-8.6	-8.6	-8.6	-7.4
External sector									
Current account balance									
(including official grants)	-9.1	-9.1	-12.2	-11.9	-9.1	-8.6	-8.6	-8.6	-7.4
(excluding official grants)	-9.7	-9.7	-12.8	-12.4	-9.4	-9.0	-9.0	-8.9	-7.6
External public debt (including IMF)	21.0	21.5	21.9	22.5	23.5	24.5	25.7	26.9	27.5
NPV of external debt outstanding	11.5	13.6	10.8	9.4	12.2	14.9	17.7	20.5	22.5
percent of exports of goods and services	30.6	36.2	26.0	26.9	36.3	46.3	57.1	67.9	71.3
Gross international reserves (millions of US\$)	5,383	5,517	5,349	4,927	5,645	6,579	7,407	8,324	10,139
Months of prospective imports of goods and services	2.9	2.9	2.8	2.5	2.7	3.0	3.2	3.3	3.8
Total donor support (millions of US\$)	1,477	1,416	1,132	1,630	1,288	1,284	1,335	1,182	1,110
percent of GDP	2.5	3.5	2.8	3.6	2.5	2.3	2.2	1.8	1.5
Central government budget									
(Percent of GDP)									
Total revenue	17.3	18.7	17.7	19.1	19.9	20.5	20.6	20.8	21.6
Grants	2.0	2.1	1.6	1.4	1.0	0.9	0.9	0.8	0.6
Total expenditure	20.7	25.0	27.7	28.0	27.5	27.4	27.6	27.7	28.3
Arrears clearance and VAT refunds	2.1	-2.2	0.1	0.9	0.9	-0.1	-0.1	-0.1	-0.1
Overall balance (financing basis)	-4.0	-6.3	-11.8	-10.0	-8.8	-6.2	-6.2	-6.2	-6.2
Net domestic financing	3.3	3.7	9.3	7.1	5.8	3.2	3.1	4.2	3.1
Central government debt (gross)	43.7	44.8	50.2	51.4	52.5	52.5	52.9	54.9	54.8
Domestic debt	22.8	21.1	28.3	29.0	29.0	28.0	27.2	28.1	27.3
External debt	21.0	23.7	21.9	22.5	23.5	24.5	25.7	26.9	27.5
Central government debt (net)	39.9	42.4	48.0	49.5	50.5	50.5	50.7	52.6	52.1
Memorandum items:									
Nominal GDP (millions of GHc)	59,816	73,919	73,109	89,689	107,335	125,800	145,743	166,573	193,204
GDP per capita (millions of U.S. dollars)	1,594	1,610	1,622	1,786	1,930	2,071	2,214	2,346	2,528

Sources: Ghanaian authorities; and Fund staff estimates and projections.

¹ Based on new national accounts rebased to 2006, including ECF program indicators. For 2018, it is assumed that new oil fields will come to stream.

² Percent of broad money (including foreign currency deposits) at the beginning of the period.

³ Including public enterprises and errors and omissions.

Table 2A. Ghana: Summary of Central Government Budgetary Operations, 2011–18

	2011	2012		2013	2014	2015	2016	2017	2018
	Prel.	6-7 ECF	Prel.			Proj.			
	(In percent of GDP)								
Total revenue and grants	19.3	20.8	19.3	20.5	21.0	21.3	21.4	21.6	22.3
Revenue	17.3	18.7	17.7	19.1	19.9	20.5	20.6	20.8	21.6
Oil revenue	1.1	1.7	1.3	1.2	1.5	1.5	1.7	1.7	2.5
Nonoil revenue	16.1	17.0	16.4	17.9	18.5	18.9	18.9	19.1	19.1
Tax revenue	15.1	15.7	15.9	17.4	18.0	18.4	18.4	18.6	18.6
Direct taxes	6.7	7.9	7.2	8.0	8.4	8.5	8.5	8.6	8.6
Indirect taxes	6.1	5.9	5.9	6.4	6.6	6.7	6.7	6.7	6.8
Trade taxes	2.5	2.8	2.7	2.9	3.0	3.1	3.2	3.3	3.2
Nontax revenue	1.0	1.2	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Total Grants	2.0	2.1	1.6	1.4	1.0	0.9	0.9	0.8	0.6
Project grants	0.9	1.1	0.9	0.9	0.6	0.5	0.6	0.6	0.4
Program grants	0.5	0.6	0.7	0.5	0.4	0.3	0.4	0.3	0.2
HIPC & MDRI assistance	0.4	0.4	0.2	0.4	0.3	0.2	0.2	0.3	0.2
Total expenditure	20.1	25.0	25.1	27.8	27.5	27.4	27.6	27.7	28.3
Recurrent expenditure	14.0	15.9	18.3	19.6	18.8	18.6	18.7	18.8	19.0
Recurrent Non-interest expenditure	11.3	12.9	15.0	16.0	15.0	14.8	14.8	14.9	14.8
Wages and salaries ¹	7.6	7.6	9.1	8.3	8.8	9.0	8.9	8.9	8.9
Goods and services	1.2	1.2	1.8	1.9	1.9	1.9	1.9	1.9	2.0
Transfers	1.9	3.1	2.6	5.0	3.1	2.7	2.8	2.8	2.8
Reserve Fund	0.6	1.0	1.5	0.8	1.1	1.1	1.2	1.2	1.2
Interest	2.7	3.0	3.3	3.6	3.8	3.9	3.9	3.9	4.2
domestic	2.2	2.3	2.6	2.9	3.1	3.3	3.1	2.9	2.9
foreign	0.5	0.7	0.8	0.7	0.7	0.6	0.7	1.0	1.3
Capital Expenditure	6.2	7.8	6.8	8.2	8.7	8.8	8.9	8.9	9.3
Domestic financed	3.3	3.4	3.3	4.2	4.7	4.8	4.9	5.0	5.3
Foreign financed	2.9	4.4	3.5	4.0	4.0	4.0	3.9	3.9	4.0
Overall balance (before arrears)	-1.0	-4.4	-6.1	-7.6	-6.6	-6.2	-6.2	-6.2	-6.2
Arrears clearance ²	-2.5	-1.9	-2.4	-2.1	-2.2	0.0	0.0	0.0	0.0
Deferred wage payments	-0.6	-1.3	-2.6	0.0	0.0	0.0	0.0	0.0	0.0
VAT refunds	-0.1	-0.1	-0.2	-0.4	-0.1	-0.1	-0.1	-0.1	-0.1
Overall balance (after arrears)	-4.1	-7.7	-11.0	-10.0	-8.8	-6.2	-6.2	-6.2	-6.2
Discrepancy	-0.1	-1.3	0.8	0.0	0.0	0.0	0.0	0.0	0.0
Financing	4.0	6.3	11.8	10.0	8.8	6.2	6.2	6.2	6.2
Divestiture receipts (net)	-1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Foreign (net)	1.4	2.4	2.4	2.7	2.9	2.9	3.1	1.9	3.0
Exceptional financing (debt relief, bilateral)	0.2	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Domestic financing (net)	3.3	3.7	9.3	7.1	5.8	3.2	3.1	4.2	3.1
Memorandum items:									
Poverty-reducing expenditures	4.8	4.6	7.3	6.0	6.2	6.5	6.5	6.5	6.5
Domestic arrears ³									
New arrears	0.5	0.0	2.1	0.0	0.0	0.0	0.0	0.0	0.0
Arrears cleared ⁴	2.3	1.9	0.6	1.3	1.0	0.0	0.0	0.0	0.0
Arrears outstanding	1.8	0.9	3.0	1.2	0.0	0.0	0.0	0.0	0.0
Total government debt (gross)	43.7	44.8	50.2	51.4	52.5	52.5	52.9	54.9	54.8
Domestic debt	22.8	21.1	28.3	29.0	29.0	28.0	27.2	28.1	27.3
External debt	21.0	23.7	21.9	22.5	23.5	24.5	25.7	26.9	27.5
Total government debt (net)	39.9	42.4	48.0	49.5	50.5	50.5	50.7	52.6	52.1

Sources: Ghanaian authorities; and IMF staff estimates and projections.

¹Excludes deferred wage payments.

²Includes clearance of liabilities of government to state-owned enterprises. For 2012, includes carry over payments from 2011 of 1.8 percent of 2012 GDP.

³Excludes government liabilities to State-owned enterprises, non-cash payments and deferred wage payments.

⁴Excludes clearance of government liabilities to state-owned enterprises, includes 0.1 percent of GDP of reconciliation and 0.3 percent of GDP of securitization.

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Table 2B. Ghana: Summary of Central Government Budgetary Operations, 2011–18

	2011	2012		2013	2014	2015	2016	2017	2018
		6-7 ECF	Prel.				Proj.		
(In millions of GHc, unless otherwise specified)									
Total revenue and grants	11,530	15,341	14,099	18,404	22,516	26,808	31,255	36,044	43,045
Revenue	10,320	13,791	12,939	17,130	21,409	25,732	29,964	34,705	41,799
Oil revenue	666	1,240	970	1,098	1,565	1,940	2,445	2,910	4,912
Nonoil revenue	9,654	12,551	11,969	16,033	19,845	23,791	27,519	31,795	36,887
Tax revenue	9,036	11,639	11,606	15,596	19,306	23,153	26,772	30,933	35,902
Direct taxes	4,037	5,874	5,266	7,210	9,032	10,753	12,400	14,269	16,588
Indirect taxes	3,667	4,324	4,350	5,767	7,067	8,486	9,759	11,219	13,093
Trade taxes	1,516	2,061	1,990	2,619	3,207	3,914	4,612	5,445	6,222
Nontax revenue	618	913	363	437	539	638	747	862	985
Total Grants	1,210	1,550	1,160	1,274	1,107	1,076	1,291	1,339	1,246
Project grants	523	794	648	812	657	689	891	926	814
Program grants	281	436	503	465	378	396	583	500	386
HIPC & MDRI assistance	242	320	146	347	279	293	308	427	428
Total expenditure	12,052	18,503	18,375	24,947	29,512	34,513	40,168	46,186	54,691
Recurrent expenditure	8,348	11,723	13,404	17,571	20,155	23,457	27,266	31,376	36,731
Recurrent Non-interest expenditure	6,737	9,531	10,968	14,386	16,055	18,566	21,598	24,815	28,665
Wages and salaries ¹	4,535	5,637	6,666	7,466	9,421	11,300	12,937	14,832	17,132
Goods and services	724	897	1,322	1,742	2,091	2,376	2,750	3,184	3,820
Transfers	1,148	2,289	1,908	4,451	3,348	3,458	4,116	4,729	5,348
Reserve Fund	330	708	1,072	727	1,195	1,433	1,795	2,070	2,365
Interest	1,611	2,191	2,436	3,185	4,100	4,890	5,668	6,561	8,065
Domestic	1,308	1,683	1,880	2,572	3,364	4,122	4,579	4,882	5,540
Foreign	303	508	556	613	736	768	1,089	1,679	2,525
Capital Expenditure	3,704	5,789	4,971	7,376	9,357	11,057	12,902	14,810	17,960
Domestic	1,963	2,508	2,437	3,751	5,018	5,987	7,206	8,394	10,292
Foreign	1,741	3,281	2,535	3,625	4,339	5,070	5,696	6,416	7,668
Overall balance (before arrears)	-611	-3,239	-4,440	-6,859	-7,122	-7,859	-9,088	-10,345	-11,882
Arrears clearance ²	-1,505	-1,430	-1,755	-1,928	-2,316	0	0	0	0
Deferred wage payments	-343	-991	-1,872	-142	0	0	0	0	0
VAT refunds	-89	-77	-164	-317	-127	-153	-175	-203	-236
Overall balance (after arrears)	-2,459	-5,660	-8,067	-8,929	-9,438	-7,859	-9,088	-10,345	-11,882
Discrepancy	-71	-991	581	0	0	0	0	0	0
Financing	2,388	4,669	8,649	8,929	9,438	7,859	9,088	10,345	11,882
Divestiture receipts (net)	-572	0	0	0	0	0	0	0	0
Foreign (net)	856	1,778	1,773	2,430	3,122	3,684	4,499	3,161	5,799
Exceptional financing (debt relief, bilateral)	116	131	44	118	113	113	108	109	98
Domestic financing (net)	1,988	2,761	6,831	6,382	6,203	4,062	4,482	7,074	5,985
Memorandum items:									
Nominal GDP (millions of GHc)	59,816	73,919	73,109	89,689	107,335	125,800	145,743	166,573	193,204
Poverty-reducing expenditures	2,883	3,434	5,350	5,366	6,623	8,198	9,497	10,855	12,590
Domestic arrears ³									
New arrears	281	0	1,561	0	0	0	0	0	0
Arrears cleared ⁴	1,362	1,430	471	1,128	1,052	0	0	0	0
Arrears outstanding	1,090	655	2,180	1,052	0	0	0	0	0
Total government debt (gross)	26,169	33,123	36,690	46,137	56,318	66,008	77,048	91,520	105,934
Domestic debt	13,622	15,572	20,715	25,968	31,120	35,182	39,663	46,738	52,723
External debt	12,547	17,551	15,975	20,168	25,198	30,826	37,385	44,782	53,211
Total government debt (net)	23,866	31,313	35,123	44,370	54,232	63,518	73,954	87,652	100,694

Sources: Ghanaian authorities; and IMF staff estimates and projections.

¹ Excludes deferred wage payments.

² Includes clearance of government liabilities to state-owned enterprises. For 2012, includes carry over payments from 2011 of GHc 1,335 million.

³ Excludes liabilities to state-owned enterprises, non-cash payments, and deferred wage payments.

⁴ Excludes liabilities to state-owned enterprises' includes GHc 48 million of reconciliation and GHc 203 million of securitization.

Table 2C. Ghana: Summary of Budgetary Central Government Operations, 2008–18
(GFS 2001)

	2011	2012	2013	2014	2015	2016	2017	2018
			Proj.	Proj.	Proj.	Proj.	Proj.	Proj.
	(In percent of GDP)							
Revenue	19.1	19.1	20.2	20.9	21.2	21.3	21.5	22.2
Taxes	15.3	16.0	17.5	18.4	18.8	19.3	19.7	20.3
Direct taxes	6.7	7.6	8.5	8.9	9.1	9.5	9.8	10.4
Personal income tax	2.3	3.0	3.2	3.4	3.5	3.5	3.5	3.5
Self-employed tax	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3
Corporate tax	2.6	3.2	3.6	3.8	3.8	3.8	3.8	3.8
Oil sector	0.0	0.0	0.1	0.1	0.2	0.7	1.0	1.5
Other direct taxes	1.6	1.1	1.2	1.3	1.3	1.2	1.2	1.2
of which: oil royalties	0.3	0.4	0.3	0.4	0.4	0.3	0.2	0.3
Indirect taxes	5.9	5.5	5.9	6.3	6.5	6.4	6.4	6.5
Excises	1.0	1.0	1.0	0.9	0.9	0.9	0.9	0.9
VAT	3.8	3.6	3.9	4.4	4.6	4.5	4.6	4.6
Communications service tax	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3
Social Contributions National Health Insurance	0.8	0.8	0.8	0.7	0.7	0.7	0.7	0.7
Trade taxes	2.5	2.7	2.9	3.0	3.1	3.2	3.3	3.2
Other tax revenues	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Other revenue	1.8	1.5	1.3	1.5	1.5	1.2	1.1	1.2
Grants	2.0	1.6	1.4	1.0	0.9	0.9	0.8	0.6
Expenditure	24.6	28.4	27.2	27.5	27.4	27.6	27.7	28.3
Expense	20.6	22.3	21.6	21.7	22.1	21.7	22.3	22.5
Compensation of employees	11.9	10.5	10.0	10.4	10.9	10.8	10.8	10.7
Wages and salaries ¹	7.6	9.1	8.3	8.8	9.0	8.9	8.9	8.9
Deferred wage payments	2.6	0.2	0.0	0.0	0.0	0.0	0.0	0.0
Social Contributions	1.7	1.2	1.7	1.6	1.9	1.9	1.9	1.8
Purchases of goods and services	1.2	1.8	1.9	1.9	1.9	1.9	1.9	2.0
Interest	2.7	3.3	3.6	3.8	3.9	3.9	3.9	4.2
Domestic ²	2.2	2.6	2.9	3.1	3.3	3.1	2.9	2.9
Foreign	0.5	0.8	0.7	0.7	0.6	0.7	1.0	1.3
Subsidies	0.0	1.9	1.6	0.4	0.0	0.0	0.0	0.1
Grants to Other Government Units ²	3.0	3.3	3.7	4.0	4.3	3.9	4.4	4.4
Other expense	1.8	1.5	0.8	1.1	1.1	1.2	1.2	1.2
Net acquisition of nonfinancial assets	4.1	5.2	5.5	5.8	5.3	5.8	5.4	5.8
Domestic financing ⁴	1.2	1.8	1.5	1.8	1.3	1.9	1.5	1.8
Foreign financing	2.9	3.5	4.0	4.0	4.0	3.9	3.9	4.0
Discrepancy	-0.1	0.8	0.0	0.0	0.0	0.0	0.0	0.0
Net lending / borrowing (overall balance)	-5.5	-9.3	-7.0	-6.6	-6.2	-6.2	-6.2	-6.2
Net financial transactions	-5.5	-9.3	-7.0	-6.6	-6.2	-6.2	-6.2	-6.2
Net acquisition of financial assets	2.1	-1.1	0.1	0.1	0.2	0.2	0.3	0.3
Currency and deposits ⁵	1.4	-1.0	0.3	0.2	0.3	0.3	0.4	0.4
Shares and other equity ⁶	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Exceptional financing (debt relief, bilateral)	-0.2	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Net incurrence of liabilities	7.6	8.3	7.1	6.7	6.4	6.4	6.5	6.5
Domestic	6.2	5.8	4.4	3.8	3.5	3.3	4.6	3.5
Debt securities	6.0	9.2	8.0	6.7	5.2	5.0	6.9	5.2
Loans	0.3	-2.9	-2.2	-1.9	-1.7	-1.7	-2.3	-1.7
Other accounts payable ⁷	-0.1	-0.4	-1.4	-1.0	0.0	0.0	0.0	0.0
Foreign	1.4	2.4	2.7	2.9	2.9	3.1	1.9	3.0
Memorandum items:								
Nominal GDP (millions of GHc)	59,816	73,109	89,689	107,335	125,800	145,743	166,573	193,204

Sources: Ghanaian authorities; and IMF staff estimates and projections.

¹ Excludes deferred wage payments which are reported on an independent line.

² Includes new arrears classified under this definition.

³ Includes cash arrears and promissory notes to statutory funds.

⁴ Includes new project-arrears.

⁵ Net transfers to Oil Fund.

⁶ Divestiture receipts (net).

⁷ Reflects net change in arrears stock (excludes government liabilities to state-owned enterprises).

GHANA

Table 2D. Ghana: Summary of Budgetary Central Government Operations, 2008–18 (GFS 2001)

	2011	2012	2013	2014	2015	2016	2017	2018
			Proj.	Proj.	Proj.	Proj.	Proj.	Proj.
(Millions of GHc, unless otherwise specified)								
Revenue	11,441	13,935	18,087	22,390	26,654	31,080	35,841	42,809
Taxes	9,131	11,712	15,663	19,726	23,678	28,109	32,744	39,209
Direct taxes	4,037	5,536	7,594	9,579	11,431	13,912	16,283	20,131
Personal income tax	1,361	2,204	2,893	3,655	4,362	5,110	5,891	6,831
Self-employed tax	132	164	248	299	372	414	478	637
Corporate tax	1,568	2,362	3,241	4,083	4,834	5,538	6,330	7,342
Oil sector	0	0	108	153	190	1,087	1,605	2,916
Other direct taxes	975	806	1,103	1,390	1,673	1,763	1,980	2,405
of which: oil royalties	184	270	277	394	488	425	409	627
Indirect taxes	3,500	4,048	5,280	6,762	8,121	9,337	10,731	12,530
Excises	606	730	904	966	1,144	1,340	1,465	1,765
VAT	2,287	2,614	3,470	4,753	5,742	6,551	7,598	8,859
Communications service tax	135	128	151	279	330	387	446	509
Social Contributions National Health Insurance	472	576	754	765	905	1,060	1,222	1,397
Trade taxes	1,516	1,990	2,619	3,207	3,914	4,612	5,445	6,222
Other tax revenues	79	138	170	179	211	248	285	326
Other revenue	1,100	1,062	1,151	1,557	1,900	1,681	1,758	2,354
Grants	1,210	1,160	1,274	1,107	1,076	1,291	1,339	1,246
Expenditure	14,710	20,752	24,361	29,512	34,513	40,168	46,186	54,691
Expense	12,339	16,336	19,416	23,272	27,863	31,681	37,201	43,541
Compensation of employees	7,106	7,683	9,004	11,163	13,693	15,692	17,996	20,595
Wages and salaries ¹	4,535	6,666	7,466	9,421	11,300	12,937	14,832	17,132
Deferred wages	1,558	142	0	0	0	0	0	0
Social Contributions	1,013	876	1,538	1,742	2,393	2,755	3,163	3,462
Purchases of goods and services	724	1,322	1,742	2,091	2,376	2,750	3,184	3,820
Interest	1,611	2,436	3,185	4,100	4,890	5,668	6,561	8,065
Domestic ²	1,308	1,880	2,572	3,364	4,122	4,579	4,882	5,540
Foreign	303	556	613	736	768	1,089	1,679	2,525
Subsidies ²	0	1,395	1,403	443	48	53	58	163
Grants to Other Government Units ²	1,820	2,428	3,355	4,281	5,423	5,722	7,333	8,533
Other expense	1,078	1,072	727	1,195	1,433	1,795	2,070	2,365
Reserve Fund	330	1,072	727	1,195	1,433	1,795	2,070	2,365
Net acquisition of nonfinancial assets	2,442	3,834	4,945	6,239	6,650	8,487	8,985	11,150
Domestic financing ⁴	701	1,300	1,320	1,900	1,580	2,792	2,569	3,482
Foreign financing	1,741	2,535	3,625	4,339	5,070	5,696	6,416	7,668
Discrepancy	-71	581	0	0	0	0	0	0
Net lending / borrowing (overall balance)	-3,269	-6,817	-6,273	-7,122	-7,859	-9,088	-10,345	-11,882
Net financial transactions	-3,269	-6,817	-6,273	-7,122	-7,859	-9,088	-10,345	-11,882
Net acquisition of financial assets	1,273	-781	108	86	204	291	480	633
Currency and deposits ⁵	817	-736	226	200	317	398	589	732
Shares and other equity ⁶	572	0	0	0	0	0	0	0
Exceptional financing (debt relief, bilateral)	-116	-44	-118	-113	-113	-108	-109	-98
Net incurrence of liabilities	4,542	6,036	6,382	7,208	8,063	9,378	10,825	12,516
Domestic	3,686	4,263	3,952	4,087	4,379	4,880	7,664	6,717
Debt securities	3,561	6,694	7,173	7,157	6,542	7,290	11,449	10,035
Loans	157	-2,105	-1,952	-2,018	-2,163	-2,410	-3,785	-3,318
Other accounts payable ⁷	-32	-326	-1,270	-1,052	0	0	0	0
Foreign	856	1,773	2,430	3,122	3,684	4,499	3,161	5,799
Memorandum items:								
Nominal GDP (millions of GHc)	59,816	73,109	89,689	107,335	125,800	145,743	166,573	193,204

Sources: Ghanaian authorities; and IMF staff estimates and projections.

¹ Excludes deferred wage payments which are reported on an independent line.

² Includes new arrears classified under this definition.

³ Includes cash arrears and promissory notes to statutory funds.

⁴ Includes new project-arrears.

⁵ Net transfers to Oil Fund.

⁶ Divestiture receipts (net).

⁷ Reflects net change in arrears stock (excludes government liabilities to state-owned enterprises).

Table 3. Ghana: Monetary Survey, 2011–18

	2011	2012	2013Q1	2013	2014	2015	2016	2017	2018	
	Prel.	6-7 ECF	Prel.	Prel.	Proj.					
(In millions of GHC, unless otherwise indicated)										
I. Depository Corporation Survey										
Net foreign assets	7,880	11,202	6,953	6,483	6,225	8,572	11,627	14,581	18,011	24,135
Net Domestic Assets	10,315	13,323	15,665	16,508	22,676	27,101	31,340	36,841	43,252	48,480
Domestic Claims	15,675	14,897	22,628	24,862	28,780	33,809	40,493	45,908	50,222	56,619
Net Claims on Central Government	5,181	2,696	7,735	9,105	9,011	10,252	11,064	11,960	13,375	13,375
Claims on Other Sectors	10,494	12,201	14,893	15,757	19,769	23,558	29,429	33,948	36,847	43,244
Claims on Other Financial Corporations	-7	0	-12	-18	0	...	...	...	...	...
Claims on State and Local Government	0	0	0	0	0	...	...	...	...	...
Claims on Public Non-financial Corporations	1,351	1,209	2,743	2,907	1,793	1,932	2,076	2,223	-6,970	-8,140
Claims on Private Sector	9,150	10,992	12,161	12,868	16,830	20,479	26,207	30,578	33,324	39,721
Other Items (Net)	-5,360	-1,574	-6,963	-8,354	-6,104	-6,708	-9,153	-9,067	-6,970	-8,140
Money and quasi-money (M3)	18,195	24,525	22,618	22,991	28,901	35,673	42,967	51,422	61,263	72,615
Broad money (M2)	14,241	18,701	17,501	17,481	21,990	26,773	31,698	37,128	43,174	49,765
Foreign exchange deposits	3,954	5,824	5,117	5,510	6,911	8,901	11,269	14,295	18,089	22,850
II. Central Bank										
Net foreign assets	6,670	8,544	5,781	5,832	5,709	7,759	10,553	13,321	16,525	22,406
Net domestic assets	-890	-1,352	2,079	1,596	4,037	4,106	3,495	3,134	2,609	-351
Net Domestic Claims	297	-1,746	3,588	3,904	6,515	6,584	5,973	5,612	5,088	2,127
Claims on Other Depository Corporations	-2,299	-5,230	-2,305	-2,640	-1,697	-1,448	81	-700	-3,817	-5,689
Net Claims on Central Government	1,943	2,761	4,140	4,695	6,382	6,203	4,062	4,482	7,074	5,985
Claims on Other Sectors ²	653	724	1,754	1,849	1,830	1,830	1,830	1,830	1,830	1,830
Other Items (Net) ³	-1,187	394	-1,509	-2,308	-2,478	-2,478	-2,478	-2,478	-2,478	-2,478
Base money	5,780	7,193	7,860	7,428	9,746	11,865	14,048	16,454	19,134	22,055
Currency In Circulation (net of cash in vaults)	3,763	4,838	4,919	4,502	6,087	7,514	9,122	11,050	13,181	15,720
Liabilities To Other Depository Corporations	1,840	2,178	2,678	2,844	3,384	4,075	4,650	5,129	5,678	6,059
Required Reserves	1,283	...	...	...	...	...	...	...	...	...
Excess Reserves	553	...	...	...	...	...	...	...	...	...
Liabilities To Other Sectors	176	176	275	268	275	275	275	275	275	275
Memorandum items:	(In 12-month percentage change; unless otherwise indicated)									
Base money	31.1	24.5	36.0	47.3	24.0	21.7	18.4	17.1	16.3	15.3
M3	32.2	34.8	24.3	24.3	27.8	23.4	20.4	19.7	19.1	18.5
Credit to the private sector	29.0	20.1	32.9	31.4	38.4	21.7	28.0	16.7	9.0	19.2
M3-to-GDP ratio (in percent)	30.4	33.2	30.9	25.3	32.2	33.2	34.2	35.3	36.8	37.6
Base money multiplier (M2/base money)	2.5	2.6	2.2	2.4	2.3	2.3	2.3	2.3	2.3	2.3
Credit to the private sector (in percent of GDP)	15.3	14.9	16.6	14.2	18.8	19.1	20.8	21.0	20.0	20.6

Sources: Ghanaian authorities; and Fund staff estimates and projections.

¹ End of period.

² Include public enterprises and the local government.

³ Including valuation.

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Table 4. Ghana: Balance of Payments, 2011–18

	2011	2012		2013	2014	2015	2016	2017	2018
		6-7 ECF	Prel.	Proj.					
(In millions of U.S. dollars)									
Current account	-3,541	-3,651	-4,924	-5,424	-4,581	-4,811	-5,251	-5,728	-5,459
Trade balance	-3,052	-2,561	-4,220	-5,279	-4,631	-4,905	-5,479	-6,011.7	-5,293
Exports, f.o.b.	12,785	14,814	13,543	13,231	14,284	15,136	15,993	17,010	19,978
Imports, f.o.b.	-15,838	-17,375	-17,763	-18,510	-18,915	-20,041	-21,472	-23,022	-25,271
Of which: oil	-3,165	-3,100	-3,331	-3,500	-3,221	-3,262	-3,341	-3,469	-3,621
Services (net)	-1,856	-1,928	-977	-1,794	-1,912	-1,872	-1,829	-1,782	-2,086
Income (net)	-1,230	-1,497	-2,132	-1,137	-997	-1,216	-1,424	-1,583	-1,525
Of which: interest on public debt	-223	-355	-224	-719	-279	-358	-468	-642	-773
Transfers	2,597	2,335	2,405	2,786	2,960	3,182	3,480	3,649	3,445
Official transfers	229	237	258	237	178	175	244	199	147
Other transfers	2,369	2,098	2,148	2,549	2,782	3,007	3,236	3,450	3,299
Capital and financial account	4,954	3,546	3,597	5,036	5,333	5,813	6,152	6,711	7,340
Capital account	445	431	283	235	212	171	168	164	164
Financial account	4,509	3,115	3,313	4,801	5,121	5,642	5,984	6,547	7,176
Foreign direct investment (net)	3,222	3,124	3,293	3,342	3,440	3,786	4,152	4,511	4,984
Portfolio investment (net)	118	210	1,122	476	506	557	427	464	513
Other investment (net)	1,169	-219	-1,102	984	1,175	1,299	1,405	1,571	1,678
Medium and long term (net)	766	916	736	1,228	1,323	1,501	1,766	1,903	2,091
Short-term (net)	404	-1,135	-1,695	-205	-148	-203	-361	-332	-413
Errors and omissions	-894	0	116	0	0	0	0	0	0
Overall balance	519	-105	-1,211	-387	752	1,002	901	983	1,880
Net international reserves (-, incr.)	-519	105	1,211	387	-752	-1,002	-901	-983	-1,880
Memorandum items:	(Percent of GDP)								
Current account	-9.1	-9.1	-12.2	-11.9	-9.1	-8.6	-8.6	-8.6	-7.4
Trade Balance	-7.9	-6.4	-10.4	-11.6	-9.2	-8.8	-9.0	-9.1	-7.2
Capital and Financial Account	12.8	8.8	8.9	11.0	10.5	10.4	10.1	10.1	10.0
Overall Balance	1.3	-0.3	-3.0	-0.8	1.5	1.8	1.5	1.5	2.6
Official grants	0.6	0.6	0.6	0.5	0.4	0.3	0.4	0.3	0.2
Gross International Reserves									
(in millions of U.S. dollars)	5,383	5,517	5,349	4,927	5,645	6,579	7,407	8,324	10,139
Months of imports	2.9	2.9	2.8	2.5	2.7	3.0	3.2	3.3	3.8

Sources: Ghanaian authorities; and Fund staff estimates and projections.

Table 5. Ghana: Financial Soundness Indicators, 2007–12

	2007	2008	2009	2010	2011	2012 Dec
Capital adequacy:						
Regulatory capital ratio	15.7	13.8	18.2	19.1	17.4	18.6
Regulatory tier 1 capital ratio	13.6	12.8	17.0	18.6	15.5	16.4
Asset quality:						
Nonperforming loans to total gross loans	6.9	7.7	16.2	17.6	14.1	13.2
Credit to total assets	50.3	52.3	43.8	40.1	37.8	42.9
Loan provision to Gross loan	4.7	6.3	9.4	9.4	7.7	6.4
Bank Provisions to NPLs	...	...	68.7	70.6	76.2	77.9
Earnings and profitability:						
Return on assets, before taxes (average)	3.3	3.7	2.8	3.8	3.9	4.8
Return on equity, before taxes (average)	3.7	23.7	2.1	2.7	2.8	3.6
Interest margin to gross income	46.1	41.3	39.4	50.1	46.8	48.5
Interest spread ¹	18.3	20.8	25.9	22.7	22.2	21.4
Liquidity:						
Core liquid assets to total assets ratio	23.3	25.2	26.3	25.3	27.8	24.1
Broad liquid assets to total assets ratio	40.7	39.4	47.2	51.3	54.9	51.0
Core liquid assets to short-term liabilities ratio	31.3	33.5	34.5	32.9	35.3	30.7
Broad liquid assets to short-term liabilities ratio	54.7	52.4	62.0	66.6	69.6	64.8
Exposure to foreign exchange risk:						
Share of foreign currency deposits in total deposits	22.3	28.4	32.7	25.4	27.4	28.9
Share of foreign liabilities in total liabilities	8.1	7.0	6.2	4.7	3.4	3.5

Source: Bank of Ghana.

¹ Average lending rate minus average (saving and demand) deposit rate.

Appendix I: Risk Assessment Matrix¹

Nature/Source of Main Threats	Likelihood of Severe Realization of Threats in the Next 1–3 Years (high, medium or low)	Impact if Realized (high, medium or low)
1. Delayed fiscal adjustment	Medium • The fiscal adjustment in 2013 falls significantly short of the budget target. • Opposition from politically important groups could make it difficult for the new government to significantly reduce current spending, such as wages and energy subsidies.	High • Stronger domestic demand could renew inflationary pressures and prompt cedi depreciation, requiring the central bank to increase already high interest rates. • High real domestic interest rates could increase the fiscal debt service, create risks to banks' loan portfolios, and reduce funding available for the private sector, hurting growth. • Continued subsidies could further undermine the financial viability of energy-sector SOEs. • Debt dynamics could worsen, further reducing capacity to handle shocks given thin fiscal buffers
2. Sudden stop or reversal of capital inflows	Medium • The large 2012 deficit was primarily financed by domestic debt, and the 2013 budget assumes a continued heavy reliance on domestic financing. • As of end-2012, 30 percent of domestic debt was held by non-residents, putting the economy at risk of a sudden stop or reversal of capital inflows.	Medium • Foreign reserve cover remains thin at less than 3 months of imports. If nonresidents decided to roll over only 50 percent of their existing bond holdings, reserves would drop by \$1.4 billion (about one month of imports). • Significant currency depreciation could reignite inflationary pressures and raise the cost of foreign debt service.

¹ The RAM shows relatively low probability events that could materially alter the baseline discussed in this report. The relative likelihood of risks listed is the staff's subjective assessment of risks surrounding this baseline.

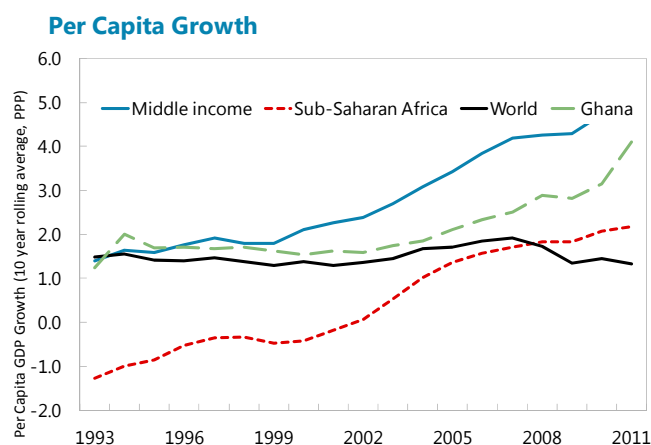
Nature/Source of Main Threats	Likelihood of Severe Realization of Threats in the Next 1–3 Years (high, medium or low)	Impact if Realized (high, medium or low)
3. Protracted slowdown in external demand and lower terms of trade	Medium - The current account deficit has widened to 12 percent of GDP, despite strong terms of trade. - A fall in commodity prices, in particular gold or cocoa, could result in a sharp contraction of exports—though part of this would be offset by lower profit and dividend payments.	Medium - A further increase in the current account deficit would weaken the already low reserve buffer. - Significant currency depreciation could reignite inflationary pressures and raise the cost of foreign debt service.
4. Spike in domestic prices	Medium Inflationary pressures are understated by temporarily low dynamics in food prices and unsustainable government subsidies that keep energy prices below market levels.	Low - Higher inflation from rising energy and food prices could require a further monetary policy tightening to prevent second-round effects, raising funding costs for the government. - A generalized increase in inflation would disproportionately affect lower income groups.

Appendix II: Inclusive Growth Successes and Challenges¹

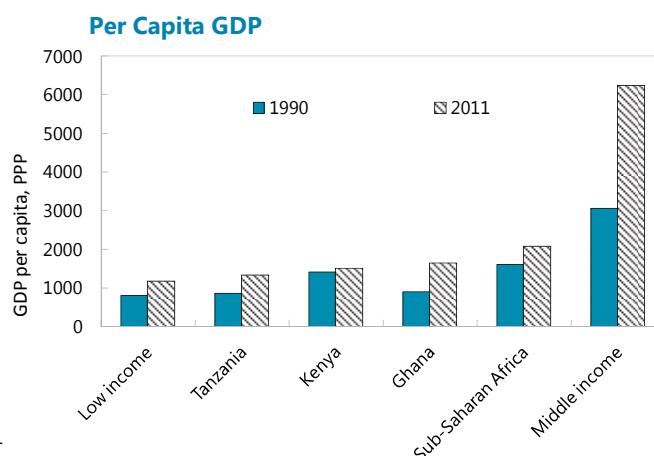
Ghana has experienced strong and broadly inclusive growth over the last 20 years, defined as growth that raises the income of most or all in society, including the poorest groups. Significant progress in poverty reduction has been achieved as well as the establishment of a highly rated governance and business environment. Growth over the medium term will depend more heavily on extractive industries, which will need to be complemented by diversified, private sector-led growth, particularly in more labor-intensive industries. This note reviews past successes and focuses on three major challenges: access to credit, energy infrastructure, and growth in labor-intensive sectors. In all three areas current macroeconomic challenges constrain the government's ability to promote robust, inclusive growth. Fiscal consolidation followed by policies targeted at these critical issues will be crucial for Ghana to advance to full middle-income status while creating 6–7 million quality jobs in the next 20 years.

Background

1. Ghana has experienced strong per capita GDP growth over the last 20 years, consistently outperforming Sub-Saharan Africa and the world. Growth has accelerated over the last 5 years, with strong performance in 2011, in particular, due to investment in oil extraction. This growth experience has recently vaulted the country into lower-middle income status. Yet over the same two decades middle-income countries as a group have experienced even stronger performance, suggesting that the government's objective of achieving full middle-income status will require sustained improvements in growth over the medium term.

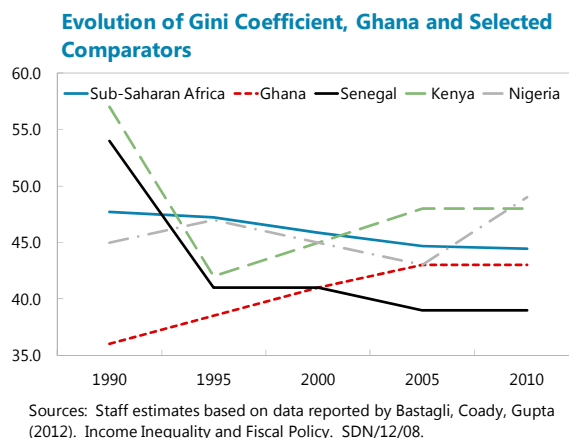
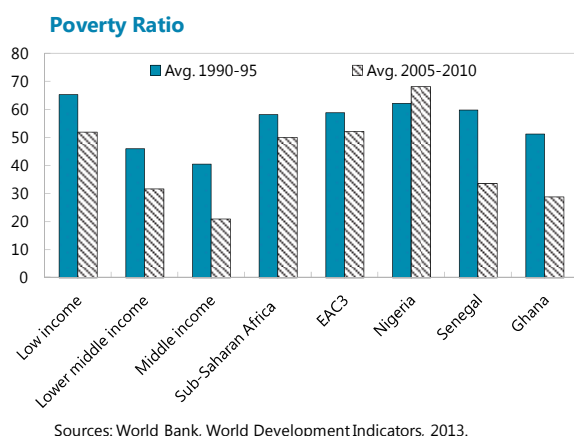


Sources: World Bank, World Development Indicators, 2013; Staff estimates.

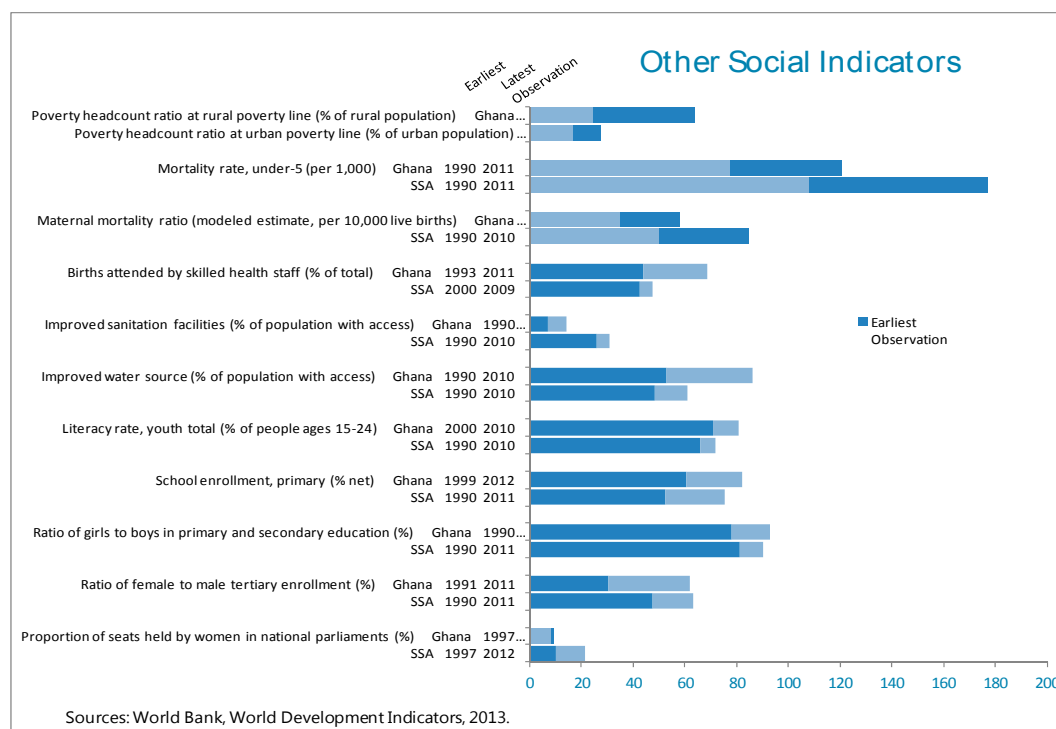


Sources: World Bank, World Development Indicators, 2013.

¹ Prepared by Kevin Wiseman.



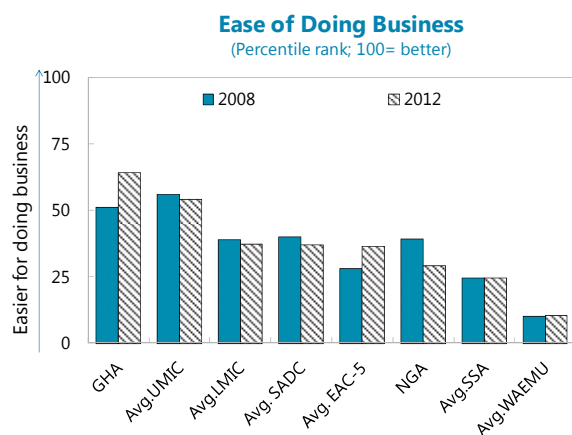
2. Ghana's period of sustained growth appears to have been widely shared. The fraction of the population living on less than \$1.25 a day has declined significantly, outperforming regional peers. Ghana has become a regional leader by poverty measures. It has significantly lower levels of poverty than the group of middle-income countries did in the early nineties, but still lags relative to those countries today. At the same time, measures of inequality have increased, with Ghana's Gini coefficient converging to regional comparators. Measures of the share of income going to the poorest 10 percent and 20 percent tell a similar story. While a rise in inequality may be considered a natural consequence of sustained growth, the experience of Senegal and Sub-Saharan Africa as a whole suggests that it is possible to reduce inequality while raising growth.



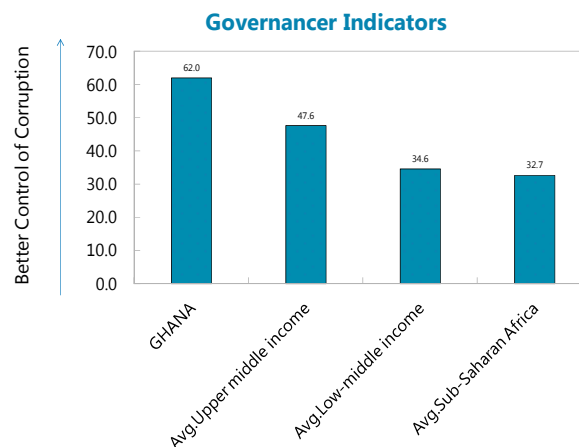
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3. Ghana's successes in poverty reduction have been also reflected in significant improvements in other Millennium Development Goal measures (Table A1). Ghana has kept pace with or outperformed the Sub-Saharan African average in mortality rates for mothers and children during births—increasingly attended by skilled health staff, in access to better water and sanitation, and in school enrollment and youth literacy. Ghana has had some success protecting particularly vulnerable populations. It has made progress in its urban/rural poverty divide from a very unequal starting point but still had the largest urban/rural consumption divide of the 6 countries studied in the Fall 2011REO. Ghana has outpaced the region in erasing gender inequality in education, and has reduced the economic activity rate of its 15–24 year olds by nearly a third between 2000 and 2006, while increasing educational opportunities, but has moved backwards in some other areas of women's equality. These achievements have been strong on balance and demonstrate that recent growth has translated into social investment and development by a variety of measures.

4. Ghana's successes in growth, poverty reduction, and quality of life improvements have been underpinned by a high-quality governance environment. Ghana has been highly rated in the World Bank's World Governance Indicators since their first observation in 1996. Its ratings have continued to improve, accelerating past even the average Upper-Middle Income Country in the World Bank's latest ease of doing business indicators.



Sources: World Bank, International Finance Corporation, Doing Business 2013



Sources: World Bank, World Governance Indicators, 2013.

Three Constraints to Inclusive Growth

A recent analysis by the U.S. and Ghanaian governments, based on firm-level surveys reporting barriers to growth, has identified access to affordable credit and reliable electricity provision as principal constraints to growth (U.S. Government and Government of Ghana, 2011). These constraints present a particularly heavy burden for small and medium-sized enterprises and labor-intensive sectors, areas where growth is most inclusive. This section analyzes these constraints and the challenge of promoting growth

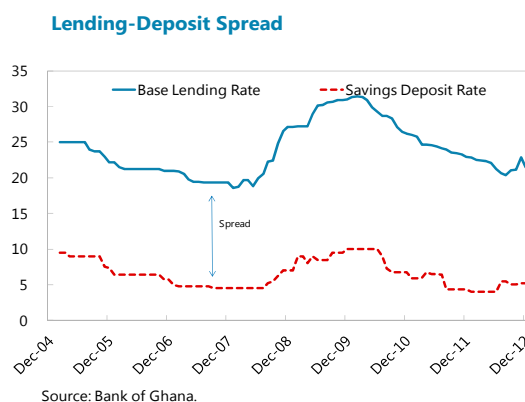
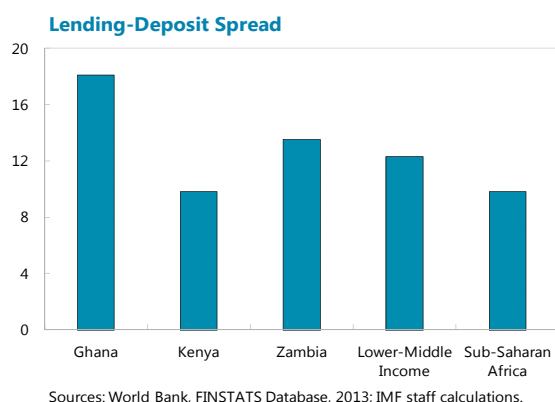
in employment-producing sectors, and suggests policies which can ensure that growth in Ghana continues to be broad-based and robust.

A. Improving Access to Credit

5. Access to affordable credit in Ghana is significantly constrained in spite of good governance indicators. The lending-deposit spread in Ghana is large relative to comparator countries and persistent over time. On one hand, the real return to savings is negative, leaving domestic savings well below that of countries with less developed financial markets. On the other hand, high real borrowing rates suppresses private sector credit which is also underdeveloped relative to similar and even less developed markets.

6. The banking system is relatively well developed, as measured by financial access. Ghana's banking system had more bank accounts and branches per capita than the regional average in 2010, and more than its income level and other structural indicators would suggest according to the World Bank's FINSTATS data. Nevertheless, it has fallen behind Kenya by these measures since 2005 and trails Nigeria by a significant margin. While Rural and Community Banks have become a key channel for financial inclusion in Ghana, only about 30 percent of adults have an account at a formal institution. A sizeable part of the population relies instead on the services of about 600 microfinance companies, as well as 3,000–5,000 individual susu collectors that serve over half a million customers.² Country comparisons of firms' access to financial services provide a similarly mixed picture; Ghana has slightly more firms with access to credit than the benchmark predicts, but fewer small firms.

7. A strong institutional ranking and increased competition have not delivered cheaper financial intermediation or reduced bank profitability. Ghana is ranked 23rd in the world and 4th in Sub-Saharan Africa in the World Bank's Access to Credit ranking in its Doing Business Indicators. The high ranking is due importantly to its establishment of a credit rating agency covering 5.7 percent of adults. According to the most recent FSAP the use of this information by lenders is in



² See Supplement on Enhancing Financial Sector Surveillance for a discussion of the different institutions, their purposes, and associated challenges.

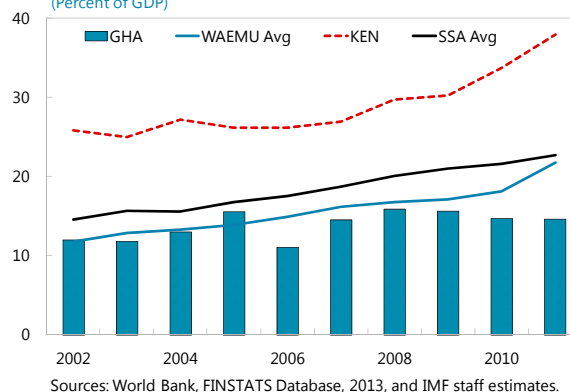
GHANA

its infancy, though the reform is likely to bear fruit over the medium term.

8. Competition is on the rise but has also failed to deliver.

There are 26 banks operating in Ghana compared to 17 in 2003, and the top three banks by size comprise less than half of banking system assets—a highly diversified sector relative to comparator countries. This competition should have already contributed to reducing the persistent lending-deposit spread and excessive bank profitability. Discussions with stakeholders suggested that the rapid growth of the financial sector has had the opposite effect—bidding up bankers' wages and increasing the cost of intermediation. These discussions suggested that the lending-deposit spread genuinely reflects operating costs in excess of banking sectors in comparator countries.

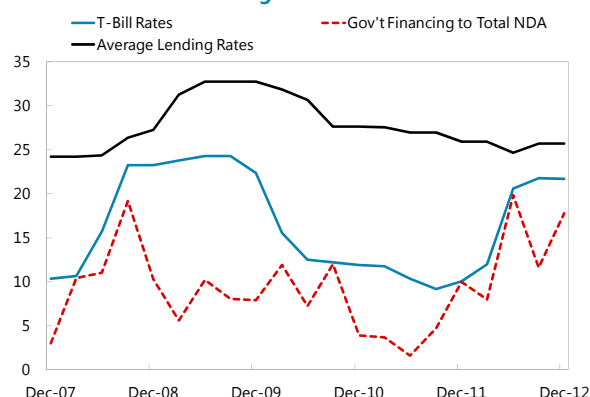
Private Credit to GDP, 2002–2011
(Percent of GDP)



9. Increases in domestic borrowing by the government are likely to further crowd out private sector credit.

A recent spike in domestic borrowing of the government has increased T-Bill rates and interbank market rates. In the past, similar increases have eventually led to rises in the average lending rates of banks. There is significant risk that short-term interest rate pressure will crowd-out credit to the private sector, particularly new lending relationships to small and medium-sized enterprises. This risks delaying recent progress in credit information coverage. Super-normal returns to government paper have led to super-normal profitability and may be preventing a rationalization of the sector.

Government Financing and Interest Rates



10. Fiscal consolidation will help alleviate difficulties in access to affordable credit. If the pattern of the spike in T-Bill rates in 2008 is repeated, a concerted fiscal consolidation will bring lending rates to lower levels over the medium term. The current above-average profitability of the banking sector should also decline as these returns to government debt normalize, likely spurring a rationalization of the crowded market for banking services. A reduction in the number of banks, concentrating assets in the hands of the more efficient banks and cooling off the labor market for seasoned bankers, should lower intermediation costs and reduce persistent lending-deposit spreads over the medium term. It will be critical that predictable and efficient bank resolution procedures are in place to respond if lower bank profitability materializes.

B. Improving the Quantity and Quality of Energy Supply

11. Ghana's infrastructure deficit, and in particular the quantity and reliability of energy supply, has been identified as another major constraint on growth. Entrepreneurs report that unreliable energy supply, the high cost of private generators, and the uncertain future of energy providers is a major concern in investment decisions. These concerns are most acute in sectors which contribute heavily to job growth—manufacturing and downstream agricultural transformation. The government faces significant operational challenges to meet its goal of generating 5,000 Mw by 2016.

12. The financial situation of energy providers is preventing necessary maintenance and additional investment. Reluctance to raise electricity prices to cost-recovery levels, along with relatively high commercial and technical losses in the distribution network, have led the state energy companies to record regular operational losses. The energy production sector is under further temporary strain from the disruption in delivery of natural gas from Nigeria through the West African Gas Pipeline, increasing the share of petroleum-based production at double the marginal cost. The pipeline is expected to resume delivery shortly, but additional energy production slated to come online is facing delays and may not be sufficient to meet growing demand. While raising electricity prices is difficult in an environment of deteriorating delivery, the sector is currently unable to maintain current infrastructure, much less ramp up investment to expand production. Shoring up the financial situation of the ECG and VRA (the state-owned distributor and producer of power, respectively) is a prerequisite to improving the operational efficiency of these bodies.

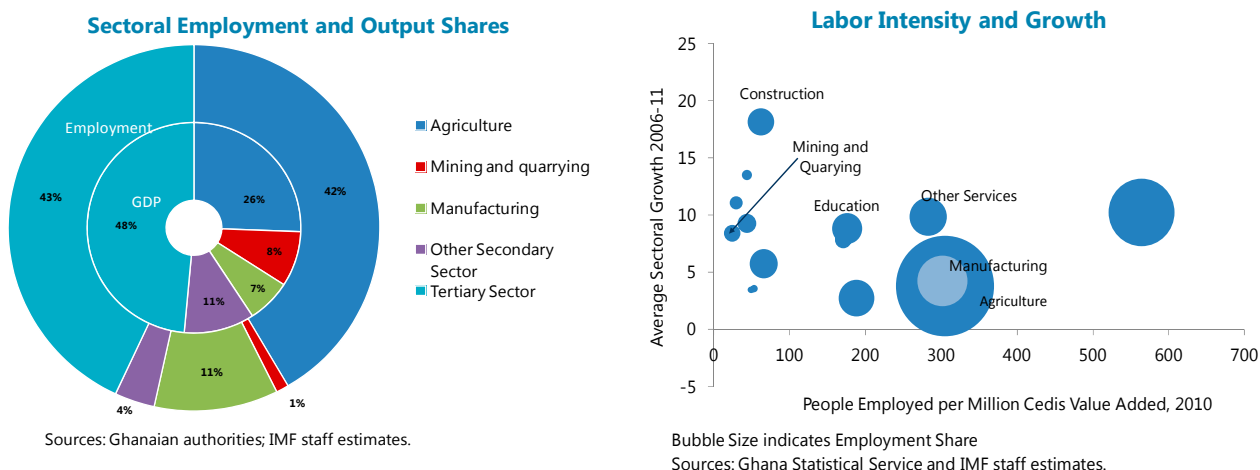
13. Further investments in new energy production will be necessary to keep pace with demand. Ghana's dynamic economy is expanding rapidly, and energy output will need to grow by an estimated 200–250 MW/year to match production growth, with more needed to improve reliability. As domestic natural gas production comes on-stream it is important that a financially viable energy sector with reduced policy uncertainty is in place to see this resource converted to electricity.

C. Growing Where the Jobs Are

14. Ghana's economy will need to add 6-7 million jobs by 2030, according to population and economic activity rate projections. Currently more than 80 percent of Ghana's jobs are in the informal sector (GSS 2012), and it is unlikely that the formal economy will be able to generate enough stable salary work to absorb the influx of new workers. More than 40 percent of workers are in the agricultural sector which remains poorly paid and largely informal. Among the middle-income SSA countries compared in the Fall 2012 REO, the agriculture share of GDP in Ghana was more than twice as high as in any other country and was declining at the second slowest rate. As discussed below, the sector has registered below-average growth over the past 5 years. While Ghana likely enjoys a natural comparative advantage in agriculture and should not pursue a manufacturing-led growth strategy at all costs, diversifying unskilled labor away from basic agriculture, while increasing

GHANA

agricultural productivity, would likely lead to higher growth—benefitting the most vulnerable within and outside the agricultural sector.



15. The recent increase in growth has been concentrated in capital-intensive sectors, limiting the positive effect on employment and inclusiveness. While labor-intensive sectors like manufacturing and agriculture underperform relative to overall growth (both under 5 percent), extractive industries, financial services, and communication have outperformed. With primary and secondary school enrollment marking significant gains over the past decade, further progress on tertiary education will be important to moving workers into these high-growth sectors. Otherwise a continuation of capital-intensive growth risks undermining Ghana's history of inclusive growth in which all benefit from greater output.

16. The extractive sector will continue to grow over the next 10 years as petroleum production peaks and new natural gas fields are brought on stream. These developments promise strong growth rates but threaten to exacerbate the capital-intensive growth of the past 5 years. Prudent use of natural resource revenues for infrastructure and human capital investments will determine whether these developments contribute positively to the welfare of the average Ghanaian worker.

17. To turn this growth dividend into jobs and raise the living standard of all Ghanaians, the government will need to leverage its high quality governance and business environment to diversify growth towards manufacturing and advanced agricultural production. While a recent AfDB analysis found that labor market regulations are a serious constraint to growth in the manufacturing sector, the evidence is mixed. Labor costs are relatively high, but those enterprises which have thrived in the current environment rate labor costs below the macroeconomic and political environment, market size, and resources as a constraint on growth. Better infrastructure, improved access to credit, further investments in health and education, and sustained macroeconomic stability will be central to ensure that the next two decades of robust growth in Ghana are as inclusive as the last two.

Table A.1. Ghana: Selected Indicators on the Millennium Development Goals

	Earliest	Earliest Year	Latest	Latest Year
Eradicate extreme poverty and hunger				
Income share held by lowest 20%	6.7	1992	5.2	2006
Malnutrition prevalence, weight for age (% of children under 5)	25.1	1993	14.3	2008
Poverty headcount ratio at \$1.25 a day (PPP) (% of population)	51.1	1992	28.6	2006
Prevalence of undernourishment (% of population)	40.5	1991	5.0	2011
Achieve universal primary education				
Literacy rate, youth female (% of females ages 15-24)	65.5	2000	79.9	2010
Literacy rate, youth male (% of males ages 15-24)	75.9	2000	81.7	2010
Persistence to last grade of primary, total (% of cohort)	62.6	1991	72.2	2008
Primary completion rate, total (% of relevant age group)	64.7	1991	99.1	2012
School enrollment, primary (% net)	60.7	1999	82.1	2012
Promote gender equality and empower women				
Proportion of seats held by women in national parliaments (%)	9.0	1997	8.3	2011
Ratio of female to male tertiary enrollment (%)	30.4	1991	61.9	2011
Ratio of female to male primary enrollment (%)	84.7	1990	94.5	2012
Ratio of female to male secondary enrollment (%)	66.2	1990	90.2	2012
Reduce child mortality				
Immunization, measles (% of children ages 12-23 months)	61.0	1990	91.0	2011
Mortality rate, infant (per 1,000 live births)	76.2	1990	51.8	2011
Mortality rate, under-5 (per 1,000 live births)	120.9	1990	77.6	2011
Improve maternal health				
Adolescent fertility rate (births per 1,000 women ages 15-19)	90.1	1997	64.1	2011
Births attended by skilled health staff (% of total)	43.8	1993	68.4	2011
Maternal mortality ratio (modeled estimate, per 100,000 live births)	580.0	1990	350.0	2010
Pregnant women receiving prenatal care (%)	85.7	1993	96.4	2011
Combat HIV/AIDS, malaria, and other diseases				
Children with fever receiving antimalarial drugs (% of children under age 5 with fever)	61.0	1998	33.0	2008
Incidence of tuberculosis (per 100,000 people)	155.0	1990	79.0	2011
Prevalence of HIV, total (% of population ages 15-49)	1.0	1990	1.5	2011
Ensure environmental sustainability				
CO2 emissions (metric tons per capita)	0.3	1990	0.3	2009
Forest area (% of land area)	32.7	1990	21.2	2011
Improved sanitation facilities (% of population with access)	7.0	1990	14.0	2010
Improved water source (% of population with access)	53.0	1990	86.0	2010
Terrestrial protected areas (% of total land area)	14.6	1990	14.7	2010
Develop a global partnership for development				
Aid per capita (current US\$)	37.8	1990	72.7	2011
Debt service (PPG and IMF only, % of exports of goods, services and primary income)	36.0	1990	3.2	2011
Internet users (per 100 people)	0.0	1990	14.1	2011
Mobile cellular subscriptions (per 100 people)	0.0	1990	84.8	2011
Telephone lines (per 100 people)	0.3	1990	1.1	2011
Other				
Fertility rate, total (births per woman)	5.6	1990	4.1	2011
GNI per capita, Atlas method (current US\$)	400	1990	1,410	2011
GNI, Atlas method (current US\$) (billions)	5.8	1990	35.1	2011
Gross capital formation (% of GDP)	14.4	1990	18.6	2011
Life expectancy at birth, total (years)	56.8	1990	64.2	2011
Literacy rate, adult total (% of people ages 15 and above)	57.9	2000	67.3	2010
Employment to population ratio, 15+, total (%)	66.2	1991	66.7	2011
Trade (% of GDP)	42.7	1990	89.2	2011

Source: World Development Indicators database, 2013.

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Appendix III: Public Investment, Growth, and Debt Sustainability¹

1. The Ghanaian authorities have emphasized the need to address large infrastructure gaps to secure Ghana's long-term growth. They intend to focus investments on productivity-enhancing infrastructure, including roads, energy infrastructure, storage facilities, and irrigation systems that would crowd in private investment.

2. While addressing infrastructure gaps is essential, the scope for debt-financed infrastructure investment is contingent on maintaining debt sustainability. Two separate tools are applied to inform the assessment of debt sustainability and the associated policy choices: the standard debt sustainability analysis (DSA) and a model-based approach. The standard DSA confirms previous results, suggesting that Ghana is able to scale up investment from around 6 to 9 percent of GDP, while maintaining a moderate risk of distress.² This conclusion holds even under conservative assumptions on growth dividends, but is contingent on successful fiscal consolidation. The second tool—and focus of this appendix—is a dynamic stochastic general equilibrium (DSGE) model that is tailored to resource-rich developing countries and explicitly captures the interaction between public investment, growth, and debt sustainability.

A. The Model

3. The DSGE model, based on Buffie et al (2012), is designed to capture many of the economic features prevalent in developing countries.³ The model suggests that a comparison of an investment's rate of return and the cost of funding is not sufficient to assess the viability of a project. Investment efficiency and absorptive capacity play an important role in determining debt sustainability. Public investment may be inefficient, that is a dollar spent on public investment does not necessarily lead to an equivalent increase in public capital, whereas absorptive capacity problems may result in implementation delays and cost overruns.

4. An additional advantage of the DSGE approach is that it specifically models the relationship between investment and growth. Not surprisingly the model predicts a positive direct relationship between public investment and growth: infrastructure investment increases the productivity of existing capital, both private and public, and spurs further investment (crowding in). At the same time, the model also captures the potentially negative effects of debt-financed public investment, whereby higher real interest rates crowd out private investment.

¹ Prepared by Genevieve Verdier and Javier Arze del Granado, based on Niangoran, A., G. Verdier and F. Zanna (2013). "Public Investment, Growth and Debt Sustainability in Ghana", mimeo.

² See supplement on Update of Joint IMF and World Bank Debt Sustainability Analysis.

³ Buffie, E., A. Berg, C. Patillo, R. Portillo and F. Zanna, (2012). "Public, Investment, Growth, and Debt Sustainability: Putting together the Pieces". IMF Working Paper WP/12/144.

GHANA

B. Application to Ghana

5. Calibrated to the Ghanaian economy, the model can match the long-run debt-to-GDP ratio projected in the standard DSA, but there are also significant differences (Table 1). First, the model suggests that projections in the DSA may underestimate the positive effect of investment on growth. Trend per capita GDP growth is assumed to be the same at around 3 percent, but the model captures the transitional effects of higher investment on economic growth. Despite higher transitional growth, the average primary balance required to achieve a similar long-term debt ratio is stronger in the model, beyond the initial scaling-up period. The reason is that higher growth in the scaling-up phase is not sufficient to generate the additional revenues needed during the repayment phase to return to the same debt ratio. These additional resource needs are met by a combination of lower current primary expenditures and higher VAT rates. The latter, in turn increases the real interest rate (via lower private savings), dampening private investment during the repayment phase.⁴

Table 1. Key Macroeconomic Variables
(In percent of non-oil GDP)

		Avg.	Avg.	
		2012	2013-18	2019-32
				2032
2013 DSA				
Real per capita GDP growth (percent)	5.2	3.6	2.7	2.8
Primary balance	-8.4	-3.5	-0.5	-0.5
Debt-to GDP ratio	51.3	54.3	55.9	53.3
Baseline (model)				
Real per capita GDP growth (percent)	...	5.4	3.6	3.3
Primary balance	...	-5.1	-0.1	-0.1
Debt-to GDP ratio	...	64.6	62.4	51.7

Source: IMF staff estimates.

6. The model results suggest that scaling up investment can be beneficial, but maintaining debt sustainability requires considerable effort on various fronts. Table 2 and Figure 1A present key results for various scenarios, suggesting the following conclusions:

Table 2. Impact of Different Scenarios
(In percent of non-oil GDP)

	Growth dividend 1/	Debt-to-GDP ratio in 2032	Increase in public capital by 2032	Increase in private capital by 2032
Baseline	15.0	51.7	32.2	10.6
Front-loaded adjustment	14.2	39.6	31.6	8.7
Gradual reform (baseline+reform)	17.6	41.1	42.6	11.3
Ambitious scaling up	20.0	68.8	41.2	13.9
Ambitious scaling up with gradual reform	23.3	48.5	54.5	15.1

1/ Difference in real per capita GDP in 2032 relative to trend (without scaling up).

Source: IMF staff estimates.

- Baseline: An increase in public investment to about 9 percent of GDP over the medium term, as projected by the authorities in the 2013 budget, can create noticeable growth dividends; real per capita income would be about 15 percent higher in 2032 than without the scaling up (growth dividend). However, the model predicts that despite the projected fiscal adjustment, the debt ratio would follow an inverted U-shape, peaking at around 70 percent of GDP in 2020, and declining to its current level (around 50 percent of GDP) by 2032. The additional investment would raise the public and private capital stocks by about 30 and 10 percent of GDP, respectively.

⁴ Note that a rising debt-servicing burden is also a feature of the standard DSA, where the debt service-to-revenue ratio approaches the established DSA threshold.

- *A front-loaded fiscal adjustment* of an additional 3 percent of GDP by 2015 (analogous to the active scenario in the DSA), would generate a slightly lower growth dividend but result in a significant decrease in the public debt ratio to less than 40 percent of GDP.
- *A more ambitious scaling up of investment* can significantly boost the growth dividend, provided there is no correlation between the size of the surge and capacity constraints. More ambitious plans, however, may jeopardize debt sustainability. The model suggests that additional investment scaling up to an average of 12 percent of GDP over the medium term would result in a debt ratio of nearly 70 percent of GDP by 2032.
- *Additional reforms to improve investment efficiency* (gradually increasing the share of public investment spending that is transformed into productive capital) coupled with an increase in user fees, would raise the growth dividend significantly, while simultaneously strengthening debt sustainability. A scenario assuming a gradual improvement of investment efficiency and user fees of a magnitude attainable over the medium term shows that the debt ratios of the baseline and ambitious investment scenarios could be reduced to about 40 and 50 percent of GDP, respectively.

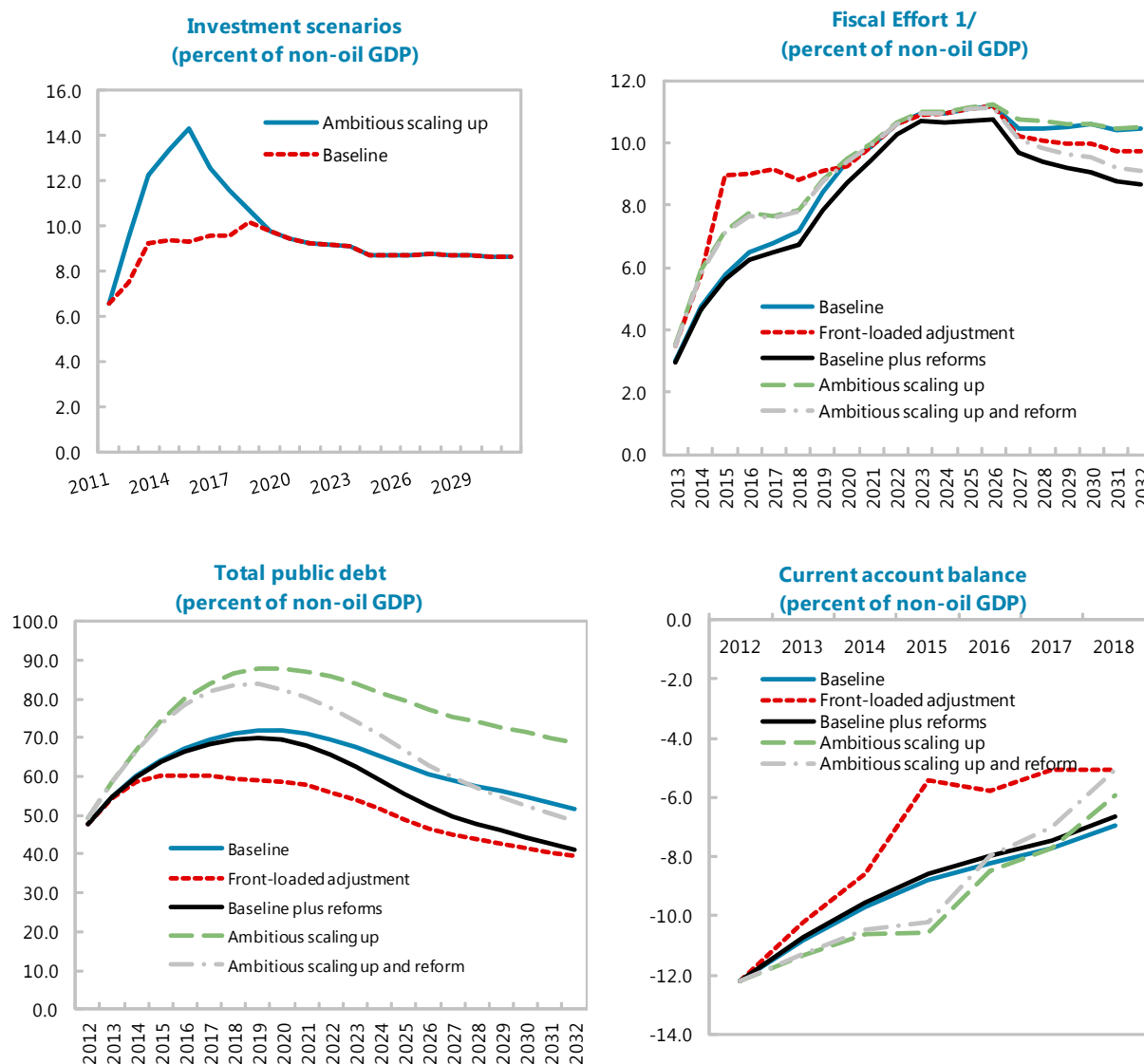
7. The different investment and reform scenarios have varied implications also for the path of the external current account adjustment. The model suggests that under the baseline scenario the assumed fiscal adjustment would lead to an improvement in the current account of almost 6 percent of GDP by 2032. More ambitious investment plans are associated with an initially smaller current account adjustment—because of the additional investment spending—but this is reversed subsequently as the investment profile adjusts to the baseline and higher tax rates are required during the repayment phase, reducing private consumption. A front-loaded fiscal consolidation would entail a significantly lower current account deficit throughout the projection period.

C. Conclusions

8. The model illustrates that a successful scaling up of public investment in Ghana will require complementary efforts to create fiscal space and improve investment efficiency. Even if the rate of return on projects is high, weak efficiency of public investment can greatly reduce the growth dividends from scaling up and increase the risk of debt distress. In addition, fiscal consolidation (significantly lower primary current spending and higher tax revenue) remains a key ingredient to debt sustainability, even with high investment efficiency; in the absence of reforms to improve the efficiency of public investment, and higher user fees, the fiscal effort required to keep the debt ratio at manageable levels would have to be larger.

GHANA

Figure 1A. Key Macroeconomic Scenarios



1/ Fiscal effort is defined as the change in the current primary deficit (i.e., excluding investment spending), net of non-tax and oil revenues) relative to its value in 2012.
Source: IMF staff estimates.

Appendix IV: Implementation of the 2011 FSAP Update—Key Policy Recommendations

The FSAP update made wide-ranging recommendations. This appendix presents a summary assessment of progress in key areas.

Banking system stability

- Most of GCB's exposure to Tema Oil Refinery was restructured. However, fiscal developments, including the re-emergence of arrears, create new concerns over government-related NPLs.
- The authorities chose not to commission an audit of problem banks, instead seeking to raise audit standards through auditor rotation and supervisory meetings with banks and auditors.
- The authorities did not establish a steering committee for bank resolution and no banks have been resolved. While there are now fewer weak public banks, there remain two.
- A Financial Stability Department has been established, training delivered (IMF TA). There are plans for a first publication soon. There has been limited progress with identifying data gaps.
- The authorities have not developed a comprehensive framework for crisis management. Powers to facilitate resolution have not been strengthened with new legislation.
- State ownership or control of the banking sector has not been reduced.

Systemic liquidity management

- Initiatives have been taken to improve the foreign exchange market (screen-based systems, BoG intervention at interbank rates, Code of Conduct), and activity has increased.
- Severe weaknesses remain in liquidity forecasting and there is a need to reform and develop the liquidity management framework and use of monetary policy instruments.

Financial sector regulation and supervision

- BoG's holdings in banks have not been reduced. For ADB, legislation is planned to change the bank's status from specialist to commercial bank, facilitating disposal of the BoG's stake.
- Forbearance remains a feature of the BoG's supervision, evidenced by its approach to weak banks and readiness to grant waivers from the large exposure (single obligor) limits.
- Plans to amend the Banking Act to strengthen the BoG's powers have not been progressed.
- Regulations have been drafted to strengthen the licensing framework, but they have not been issued, nor have guidelines to enhance industry risk management.

GHANA



INTERNATIONAL MONETARY FUND

Public Information Notice

EXTERNAL
RELATIONS
DEPARTMENT

Appendix V: Draft Public Information Notice

Public Information Notice (PIN) No. 13/xx
FOR IMMEDIATE RELEASE
[June 12, 2013]

International Monetary Fund
700 19th Street, NW
Washington, D. C. 20431 USA

IMF Executive Board Concludes Article IV Consultation with Ghana

On June 12, 2013, the Executive Board of the International Monetary Fund (IMF) concluded the Article IV consultation with Ghana.¹

Background

Economic growth continued at a robust pace amid rising fiscal and external imbalances. 2012 GDP growth of close to 8 percent was supported by expansionary fiscal policy, mirroring patterns of past election years. Fiscal pressures came to the fore in a rising public sector wage bill and costly energy subsidies that pushed the deficit close to 12 percent of GDP. The fiscal expansion also led to a significant deterioration in the public debt ratio and a widening deficit in the external current account.

¹ Under Article IV of the IMF's Articles of Agreement, the IMF holds bilateral discussions with members, usually every year. A staff team visits the country, collects economic and financial information, and discusses with officials the country's economic developments and policies. On return to headquarters, the staff prepares a report, which forms the basis for discussion by the Executive Board. At the conclusion of the discussion, the Managing Director, as Chairman of the Board, summarizes the views of Executive Directors, and this summary is transmitted to the country's authorities. An explanation of any qualifiers used in summings up can be found here: <http://www.imf.org/external/np/sec/misc/qualifiers.htm>.

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The policy mix was suboptimal. While fiscal policy became increasingly expansionary in the course of 2012, the Bank of Ghana (BoG) tightened monetary policy in the second half of the year to arrest a rapid depreciation of the cedi. The currency subsequently stabilized, with recent depreciations in line with inflation differentials, but at the cost of double-digit real interest rates. Although consumer price inflation stayed in the single digits in 2012, this outcome was helped by low food and repressed domestic fuel prices. With rising core inflation (excluding food and energy) and a February increase in fuel prices, inflation has moved back above 10 percent.

The growth momentum continues into 2013, with inflation on the rise. While non-oil growth is likely to decelerate, as a result of energy disruptions and high real interest rates, increased oil production is projected to keep overall growth close to 8 percent. Survey-based inflation expectations remain elevated at above 10 percent. The current account deficit is projected to stay high at 12 percent of GDP, despite a moderation in import growth, reflecting a weaker outlook for cocoa and gold exports. Staff projects a small reduction in the fiscal deficit (financing) to 10 percent of GDP this year, about 1 percent of GDP higher than the authorities' budget projections, mainly reflecting higher cost of energy subsidies.

A reduction in macroeconomic imbalances over the medium-term is contingent on strengthened policies. Non-oil growth is projected to stabilize at a still robust level of 5–6 percent, and inflation should gradually decline as policies are rebalanced. A planned reduction in the fiscal deficit to about 6 percent of GDP is feasible by 2015, if measures are implemented as envisaged. This, together with increased oil and gas production from new fields, would reduce the current account deficit to a still high level of 7½ percent of GDP by 2018, financed in part by strong foreign direct investment (FDI).

GHANA

Short-term stability risks are significant. Deteriorating public debt dynamics and low official reserve buffers make the economy susceptible to risks from delayed fiscal adjustment, weaker terms of trade, and rollover risks from foreign holdings of government bonds. Domestically, energy sector problems pose downside risks to growth and fiscal outcomes, while high interest rates spell risks of a renewed deterioration in banks' credit portfolios.

Executive Board Assessment

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Public Information Notices (PINs) form part of the IMF's efforts to promote transparency of the IMF's views and analysis of economic developments and policies. With the consent of the country (or countries) concerned, PINs are issued after Executive Board discussions of Article IV consultations with member countries, of its surveillance of developments at the regional level, of post-program monitoring, and of ex post assessments of member countries with longer-term program engagements. PINs are also issued after Executive Board discussions of general policy matters, unless otherwise decided by the Executive Board in a particular case.

Table 1. Ghana: Selected Economic and Financial Indicators, 2010-13¹

	2010	2011	2012	2013
	Act.	Act.	Est.	Proj.
(Annual percent change ; unless otherwise specified)				
National account and prices				
GDP at constant prices ¹	8.0	15	7.9	7.9
Real GDP (nonoil)	6.5	9.4	7.8	5.9
Real GDP per capita	5.3	12.1	5.2	5.2
GDP deflator	16.5	13	13.3	13.7
Consumer prices				
Consumer price index (annual average)	107	8.7	9.2	10.3
Consumer price index (end of period)	8.6	8.6	8.8	10.8
Money and credit				
Net domestic assets ²	27.1	28.7	51.9	44.8
Credit to the private sector ²	25.7	29	32.9	38.4
Broad money (M3, including foreign currency deposits)	34.6	32.2	24.3	27.8
Velocity (GDP/M2, end of period)	3.3	3.1	3	2.9
Base money	45.0	31.1	36	24
Banks' lending rate (weighted average; percent)	...	25.9	25.7	...
Policy rate (in percent, end of period)	13.5	12.5	15	...
(Percent of GDP)				
External sector				
Current account balance				
(including official grants)	-8.6	-9.1	-12.2	-11.9
(excluding official grants)	-9.2	-9.7	-12.8	-12.4
External public debt (including IMF)	20.0	21	21.9	22.5
NPV of external debt outstanding	8.4	11.5	10.8	9.4
percent of exports of goods and services	28.6	30.6	26	26.9
Gross international reserves (mn. of US\$)	4,680	5,383	5,349	4,927
Months of prospective Imp. of goods services	2.9	2.9	2.8	2.5
Total donor support (millions of US\$)	1,595	1,477	1,132	1,630
percent of GDP	3.5	2.5	2.8	3.6
Central government budget				
Total revenue	14.5	17.3	17.7	19.1
Grants	2.4	2	1.6	1.4
Total expenditure	22.8	20.7	27.7	28
Arrears clearance and VAT refunds	1.1	2.1	0.1	0.9
Overall balance (financing basis)	-7.2	-4	-11.8	-10
Net domestic financing	4.7	3.3	9.3	7.1
Central government debt (gross)	46.4	43.7	50.2	51.4
Domestic debt	26.3	22.8	28.3	29
External debt	20.1	21	21.9	22.5
Central government debt (net)	43.2	39.9	48	49.5
Memorandum items:				
Nominal GDP (millions of GHc)	46,043	59,816	73,109	89,689
GDP per capita (millions of U.S. dollars)	1,358	1,594	1,622	1,786

Sources: Ghanaian authorities; and IMF staff estimates and projections

¹ Based on new national accounts rebased to 2006, including ECF program indicators. For 2018, it is assumed that new oil fields will come to stream.

² Percent of broad money (including foreign currency deposits) at the beginning of the period.